

Machine Learning with Applications in Neural Networks

Lecture Notes

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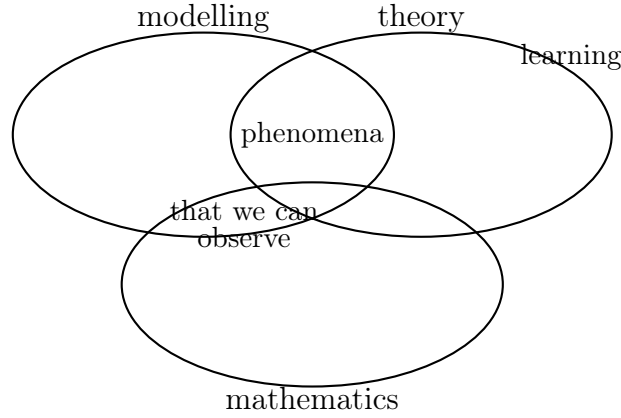
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1 Introduction and Basic Concepts

When choosing between 2 models, it is better to choose the simpler one — precisely because of over-complication we get *overfitting*.



See [Vap99]: ch. 1 (pp. 1–20) — formulation of the learning problem; [AB99]: ch. 1–2.

Basic objects:

- X — a given set
- Y — something we can observe and want to predict based on X
- We seek some function $X \rightarrow Y$
- Λ — a set of parameters
- $h : \Lambda \times X \rightarrow Y$
- Λ parametrizes a set of models $h(\lambda, \cdot) : X \rightarrow Y$, $\lambda \in \Lambda$
- $\ell : Y \times Y \rightarrow \mathbb{R}_+ = [0, \infty)$ — ℓ measures the distance between points of Y

We can observe $(x, y) \in X \times Y$. On the other hand we can consider $\lambda \in \Lambda$ and $h(\lambda, x)$ — the result of our prediction. $\ell(h(\lambda, x), y)$ — how accurate the prediction of λ (at x) is.

We assume that on $Z = X \times Y$ there is a distribution P , which we do not know.

- We would like to find $\lambda^* \in \Lambda$:

$$\rho(\lambda^*) = \min_{\lambda} \rho(\lambda'), \quad \rho(\lambda) = \mathbb{E}_z[\ell(h(\lambda, x), y)]$$

- We have data $(x_i, y_i) \stackrel{i.i.d.}{\sim} P$ (“independently and identically distributed according to P ”)
- $Z = (x_i, y_i)$. We are given a sequence $\bar{Z} = (z_1, \dots, z_m)$. $L(\bar{Z}) = \lambda \in \Lambda$?

We can compare $L(\bar{Z}) = \lambda$ with λ^* via $\rho(L(\bar{Z})) - \rho(\lambda^*)$:

$$P^m(\rho(L(\bar{Z})) - \rho(\lambda^*) > \varepsilon) < \delta, \quad \varepsilon, \delta > 0$$

Definition 1.1 (Machine Learning Algorithm): Let X, Y, Λ, h, ℓ be as above. $Z^* = (X \times Y)^m$. We call a function $L : Z^* \rightarrow \Lambda$ a *machine learning algorithm* if for $\forall \varepsilon > 0, \forall \delta > 0$ there exists $m_0 = m_0(\varepsilon, \delta)$, such that for every $m > m_0$ and every P on Z :

$$P^m(\rho(L(\bar{Z})) - \rho(\lambda^*) > \varepsilon) < \delta$$

where ε is the admissible error and δ is the probability of error.

$$\rho(\lambda) = \mathbb{E}_{(x,y) \sim P}[\ell(h(\lambda, x), y)], \quad \rho(\lambda^*) = \min_{\lambda \in \Lambda} \rho(\lambda) \quad \text{or} \quad \rho^* = \inf_{\lambda} \rho(\lambda)$$

“The problem is when we do not know P .”

1.1 Algorithm for Finite Λ for a “Classifier”

Assumptions:

$$\Lambda\text{--finite}, \quad \ell(y_1, y_2) = \begin{cases} 0, & y_1 = y_2 \\ 1, & y_1 \neq y_2 \end{cases}, \quad Y = \{0, 1\}, \quad h : \Lambda \times X \rightarrow \{0, 1\}, \quad Z = X \times \{0, 1\}$$

Empirical error: $\hat{\rho} : \Lambda \times Z^* \rightarrow \mathbb{R}_+$

$$\hat{\rho}(\lambda, \bar{Z}) = \frac{1}{m} \sum_{i=1}^m \ell(h(\lambda, x_i), y_i), \quad \bar{Z} = (z_1, \dots, z_m), \quad z_i = (x_i, y_i)$$

$$\hat{\lambda} = \operatorname{argmin}_{\lambda \in \Lambda} \hat{\rho}(\lambda, \bar{Z}), \quad \hat{\lambda} = \hat{\lambda}(\bar{Z}) \quad (*)$$

$\rightarrow \lambda^*$ is the optimal value without data; given $\bar{Z}$ — i.e. the theoretical algorithm.

Proposition 1.1 (Proposition 1): Under the assumptions: Λ —finite, $Y = \{0, 1\}$, $\ell(y_1, y_2) = \mathbb{1}_{y_1 \neq y_2}$, $L(\bar{Z}) = \hat{\lambda}$ is a machine learning algorithm.

Remark 1.2: Intuitively: for finite Λ each λ behaves like an independent coin — the law of large numbers guarantees that the empirical mean converges to the true one. The union bound over all $\lambda \in \Lambda$ adds only a logarithmic factor $\ln |\Lambda|$ to the required number of examples m . Practically: if Λ grows polynomially with d , then $m \sim d$ examples suffice — a linear dependence, which is good.

See [Vap99]: ch. 3, pp. 69–89 (basic inequalities); [MU05]: ch. 4, pp. 66–100 (Hoeffding and Chernoff bounds).

Idea 1: The goal is $A = \{\bar{Z} \in Z^* \mid \rho(\hat{\lambda}) - \rho(\lambda^*) > \varepsilon\}$, $P(A) \leq ?$

$$\rho(\hat{\lambda}) - \rho(\lambda^*) = \underbrace{\rho(\hat{\lambda}) - \hat{\rho}(\hat{\lambda}, \bar{Z})}_{(I)} + \underbrace{\hat{\rho}(\hat{\lambda}, \bar{Z}) - \hat{\rho}(\lambda^*, \bar{Z})}_{\leq 0 \text{ (by definition)}} + \underbrace{\hat{\rho}(\lambda^*, \bar{Z}) - \rho(\lambda^*)}_{(II)}$$

Idea 2:

$$B_\varepsilon = \left\{ \exists \hat{\lambda} \in \Lambda : \left| \hat{\rho}(\hat{\lambda}, \bar{z}) - \hat{\rho}(\lambda^*, \bar{z}) \right| > \frac{\varepsilon}{4} \right\}$$

From part 1 ($A_\varepsilon \subseteq B_\varepsilon$) $\Rightarrow P(A_\varepsilon) \leq P(B_\varepsilon)$.

$$C_\varepsilon = \left\{ \max_{\lambda \in \Lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{4} \right\}$$

If $\bar{z} \in B_\varepsilon$, then $|\hat{\rho}(\hat{\lambda}, \bar{z}) - \hat{\rho}(\lambda^*, \bar{z})| > \frac{\varepsilon}{4}$. By the pigeonhole principle:

$$|\rho(\hat{\lambda}) - \hat{\rho}(\hat{\lambda}, \bar{z})| > \frac{\varepsilon}{8} \quad \text{or} \quad |\rho(\lambda^*) - \hat{\rho}(\lambda^*, \bar{z})| > \frac{\varepsilon}{8}$$

Since $\hat{\lambda}, \lambda^* \in \Lambda$:

$$\max_{\lambda \in \Lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{8} \Rightarrow \bar{z} \in C_\varepsilon$$

Hence $B_\varepsilon \subseteq C_\varepsilon$, i.e. $P(A_\varepsilon) \leq P(B_\varepsilon) \leq P(C_\varepsilon)$.

Idea 3:

$$P(C_\varepsilon) = P\left(\bigcup_{\lambda \in \Lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{4}\right) \leq |\Lambda| \max_{\lambda \in \Lambda} P\left(|\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{4}\right)$$

1.2 Hoeffding's Inequality

Theorem 1.3 (Hoeffding's Inequality): Let $X_1, \dots, X_n$ be independent random variables with $a_i \leq X_i \leq b_i$ ($a_i, b_i \in \mathbb{R}$). Then for every $t > 0$:

$$P\left(\sum_{i=1}^n (X_i - \mathbb{E}X_i) > t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Remark 1.4: Inaccuracy in the notes: Hoeffding's inequality in its original form holds for a one-sided deviation $P(\sum (X_i - \mathbb{E}X_i) > t)$; the two-sided version $P(|\cdot| > t) \leq 2 \exp(\dots)$ follows by applying the inequality twice (for $+t$ and $-t$). The version in the notes is two-sided — correct, but requires a factor of 2. **Practically:** Hoeffding gives *sub-Gaussian* decay — the probability of a large deviation falls like e^{-ct^2} , which is much faster than the polynomial decay of Markov. This is the foundation of PAC-learning: it suffices to have $m = O(\frac{1}{\varepsilon^2} \ln \frac{1}{\delta})$ examples.

See [Vap99]: ch. 3, pp. 70–72; [MU05]: ch. 4 (pp. 66–100), Theorem 4.16 (Hoeffding); [AB99]: ch. 6.

In our case: $X_i = \mathbb{1}_{h(\lambda, x_i) \neq y_i}$, $(x_i, y_i) = z_i$ and $\bar{z} = (z_1, \dots, z_m)$. $\Rightarrow X_i \in [0, 1]$ and accordingly $a_i = 0 \leq X_i \leq 1 = b_i$.

$$\hat{\rho}(\lambda, \bar{z}) - \rho(\lambda) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(\lambda, x_i) \neq y_i} - \mathbb{E} \mathbb{1}_{h(\lambda, x) \neq y}$$

Now from Hoeffding's inequality:

$$P\left(|\hat{\rho}(\lambda, \bar{z}) - \rho(\lambda)| > \frac{\varepsilon}{4}\right) \leq 2 \exp\left(-\frac{2(m^2 \varepsilon^2 / 16)}{m}\right) = 2 \exp\left(-\frac{m \varepsilon^2}{8}\right)$$

independently of λ .

Idea 4: From Idea 3:

$$P(A_\varepsilon) \leq |\Lambda| \max_{\lambda \in \Lambda} P\left(|\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{4}\right) \leq 2|\Lambda| \exp\left(-\frac{m\varepsilon^2}{8}\right)$$

Applications:

1. If we want $P(A_\varepsilon) \leq \delta$, it suffices that $2|\Lambda| \exp\left(-\frac{m\varepsilon^2}{8}\right) \leq \delta$:

- Fixed ε : $m \geq \frac{8}{\varepsilon^2} \ln \frac{2|\Lambda|}{\delta}$
- Fixed m : $\varepsilon \geq \sqrt{\frac{8}{m} \ln\left(\frac{2|\Lambda|}{\delta}\right)}$
- For example if $\delta = \frac{1}{10}$: $\varepsilon = \sqrt{\frac{8}{m} \ln(20|\Lambda|)}$

2. If ε, m are fixed, then with probability $> 1 - 2|\Lambda| \cdot \exp\left(-\frac{m\varepsilon^2}{8}\right)$: $|\hat{\rho}(\lambda) - \rho(\lambda)| < \varepsilon$.

More concretely, if $\mathcal{X} = \mathbb{R}^d$, $\Lambda = \mathbb{R}^d \times \mathbb{R}$,

$$h(\langle \omega, \theta \rangle, x) = \begin{cases} 1, & \langle \omega, x \rangle - \theta \geq 0 \\ 0, & \langle \omega, x \rangle - \theta < 0 \end{cases}$$

The result would hold if $\Lambda \subseteq \mathbb{R}^d \times \mathbb{R}$.

1.3 Markov's Inequality and the Cramér–Chernoff Method

Theorem 1.5 (Markov's Inequality): Let X be a nonnegative random variable with finite expectation. Let $a > 0$. Then

$$P(X > a) \leq \frac{\mathbb{E}X}{a}$$

Proof. $\mathbb{E}X = \mathbb{E}X \cdot \mathbb{1}_{X>a} + \mathbb{E}X \cdot \mathbb{1}_{X \leq a} \geq \mathbb{E}a \cdot \mathbb{1}_{X>a} = a \cdot P(X > a) \Rightarrow P(X > a) \leq \frac{\mathbb{E}X}{a}$. $\square$

1.4 The Cramér–Chernoff Method

X — a random variable, $t \in \mathbb{R}$. We seek $P(X > t)$?

$$\begin{aligned} X > t &\iff \lambda X > \lambda t \iff e^{\lambda X} > e^{\lambda t} \\ P(X > t) &\leq P(e^{\lambda X} > e^{\lambda t}) \stackrel{\text{Markov}}{\leq} \frac{\mathbb{E}e^{\lambda X}}{e^{\lambda t}}, \quad \lambda > 0 \end{aligned}$$

1.5 The Function Ψ_X and Proof of Hoeffding's Inequality

Let X be a random variable with $\mathbb{E}X = 0$ and $\mathbb{E}e^{\lambda X} < \infty$ for $\lambda > 0$. Consider $\Psi_X(\lambda) = \log \mathbb{E}e^{\lambda X}$.

Then from the Cramér–Chernoff method:

$$P(X > t) \leq \frac{\exp(\Psi_X(\lambda))}{\exp(\lambda t)} = \exp(\Psi_X(\lambda) - \lambda t)$$

Proposition 1.6: Let X be a bounded random variable ($a \leq X \leq b$) with $\mathbb{E}X = 0$. Then:

1. $\Psi_X(0) = 0$
2. $\frac{d}{d\lambda}\Psi_X(0) = 0$
3. $\frac{d^2}{d\lambda^2}\Psi_X(\lambda) \leq \frac{1}{4}(b-a)^2$
4. Corollary: $\Psi_X(\lambda) \leq \frac{\lambda^2}{8}(b-a)^2$

Proof. 1. $\Psi_X(0) = \log \mathbb{E}e^0 = \log 1 = 0$

2.

$$\begin{aligned} \frac{d}{d\lambda}\Psi_X(\lambda) &= \frac{d}{d\lambda} \log \mathbb{E}e^{\lambda X} = \frac{1}{\mathbb{E}e^{\lambda X}} \cdot \mathbb{E}\left(\frac{d}{d\lambda}e^{\lambda X}\right) = \frac{\mathbb{E}Xe^{\lambda X}}{\mathbb{E}e^{\lambda X}} \\ \frac{d}{d\lambda}\Psi_X(0) &= \frac{\mathbb{E}(X \cdot 1)}{\mathbb{E}(1)} = \mathbb{E}X = 0 \end{aligned}$$

3.

$$\frac{d^2}{d\lambda^2}\Psi_X(\lambda) = \frac{\mathbb{E}X^2e^{\lambda X} \cdot \mathbb{E}e^{\lambda X} - (\mathbb{E}Xe^{\lambda X})^2}{(\mathbb{E}e^{\lambda X})^2} = \frac{\mathbb{E}X^2e^{\lambda X}}{\mathbb{E}e^{\lambda X}} - \left(\frac{\mathbb{E}Xe^{\lambda X}}{\mathbb{E}e^{\lambda X}}\right)^2$$

Let $g(x) = \frac{e^{\lambda X}}{\mathbb{E}e^{\lambda X}} > 0$, $\mathbb{E}g(X) = 1$. Then

$$\frac{d^2}{d\lambda^2}\Psi_X(\lambda) = \mathbb{E}X^2g(X) - (\mathbb{E}Xg(X))^2$$

Special case $g(x) = 1$:

$$\mathbb{E}X^2 - (\mathbb{E}X)^2 = \mathbb{E}(X - \mathbb{E}X)^2 = \min_c \mathbb{E}(X - c)^2 \leq \mathbb{E}\left(X - \frac{b+a}{2}\right)^2 \leq \frac{(b-a)^2}{4}$$

(this is precisely the variance — the mean square error).

In the general case, since $g(x) \geq 0$, $\mathbb{E}g(X) = 1$, we consider the distribution Q : $Q(A) = \mathbb{E}g(X)\mathbb{1}_A$. At $c = \frac{b+a}{2}$:

$$\mathbb{E}X^2g(X) - (\mathbb{E}Xg(X))^2 \leq \mathbb{E}\left(X - \frac{a+b}{2}\right)^2 g(X) \leq \left(\frac{b-a}{2}\right)^2 \mathbb{E}g(X) = \frac{(b-a)^2}{4}$$

4. From (1), (2), (3) we perform numerical differentiation and expand $\Psi_X(\lambda)$ in a Taylor series around 0:

$$\Psi_X(\lambda) = \Psi_X(0) + \lambda \Psi'_X(0) + \frac{\lambda^2}{2} \Psi''_X(\theta) \leq 0 + 0 + \frac{\lambda^2}{2} \cdot \frac{(b-a)^2}{4} = \frac{\lambda^2(b-a)^2}{8}$$

□

Remark 1.7: If $X_1, \dots, X_n$ are independent, then $X = \sum X_i$ and

$$\Psi_X(\lambda) = \log \mathbb{E} e^{\lambda X} = \log \mathbb{E} e^{\lambda \sum X_i} = \log \prod \mathbb{E} e^{\lambda X_i} = \sum \Psi_{X_i}(\lambda)$$

1.6 Proof of Hoeffding's Inequality

1. $Y_i = X_i - \mathbb{E}X_i$ (centering):

- $\mathbb{E}Y_i = 0$
- $Y_i \in [a_i - \mathbb{E}X_i, b_i - \mathbb{E}X_i] = [a'_i, b'_i]$, with $b'_i - a'_i = b_i - a_i$
- From the previous proposition: $\Psi_{Y_i}(\lambda) \leq \frac{\lambda^2(b_i - a_i)^2}{8}$

2. $Y = \sum Y_i = \sum (X_i - \mathbb{E}X_i)$:

$$P\left(\sum (X_i - \mathbb{E}X_i) > t\right) = P(Y > t) \leq \frac{\mathbb{E} e^{\lambda Y}}{e^{\lambda t}} = \exp\left(\sum \Psi_{Y_i}(\lambda) - \lambda t\right)$$

- 3.

$$\sum \Psi_{Y_i}(\lambda) \leq \frac{(b_i - a_i)^2 \lambda^2}{8} \Rightarrow P\left(\sum (X_i - \mathbb{E}X_i) > t\right) \leq \exp\left(\frac{\sum (b_i - a_i)^2 \lambda^2}{8} - \lambda t\right)$$

The optimal $\lambda^* = \frac{4t}{\sum (b_i - a_i)^2}$:

$$P\left(\sum (X_i - \mathbb{E}X_i) > t\right) \leq \exp\left(-\frac{2t^2}{\sum (b_i - a_i)^2}\right)$$

2 Measurable Functions

$(\Omega, \mathcal{F})$ measurable; $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ Borel sigma-algebra.

$X : \Omega \rightarrow \mathbb{R}$ is “measurable” (called a random variable) if $X^{-1}(A) \in \mathcal{F}$ for $\forall A \in \mathcal{B}(\mathbb{R})$.

$$\sigma(X) = \sigma(X^{-1}(A) \mid A \in \mathcal{B}(\mathbb{R}))$$

Diagram:

$$(E, \mathcal{E}) \xrightarrow{f} (\Omega, \mathcal{F}) \xrightarrow{g} (\mathbb{R}, \mathcal{B}(\mathbb{R})), \quad h = g \circ f$$

$$\mathcal{E} = \sigma(f^{-1}(A) \mid A \in \mathcal{E})$$

Then every r.v. $h : E \rightarrow \mathbb{R}$ is a composition $h = f \circ g$ for some r.v. $g : \Omega \rightarrow \mathbb{R}$ (**Doob–Dynkin Theorem**).

$\Rightarrow$ The theorem tells us that Ω is irrelevant — we can have r.v.’s without explicitly defining Ω .

2.1 Examples of Random Variables

1. $A \in \mathcal{F}$, $\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$. Then $\mathbb{1}_A^{-1}((a, b])$ is: $\emptyset$ (if $0, 1 \notin (a, b]$), A (if $1 \in (a, b]$, $0 \notin (a, b]$), Ω (if $0, 1 \in (a, b]$), $A^c = \Omega \setminus A$ (if $0 \in (a, b]$, $1 \notin (a, b]$).
2. If $A \in \mathcal{F}$, $c \in \mathbb{R}$, then $c\mathbb{1}_A$ is measurable: $(cf)^{-1}((a, b]) = f^{-1}((a/c, b/c])$ for $c > 0$.
3. If $A_1, \dots, A_n$ are disjoint sets from $\mathcal{F}$ and $c_1, \dots, c_n \in \mathbb{R}$, then $f(\omega) = \sum c_i \mathbb{1}_{A_i}(\omega)$ is measurable; $f^{-1}((a, b]) = \bigcup A_k$.
4. $f : \Omega \rightarrow \mathbb{R}$ is measurable and $0 \leq f(\omega) \leq N$. For $n \in \mathbb{N}$, $n > 0$:

$$A_{k,n} = \left\{ \omega \mid \frac{k-1}{n} < f(\omega) \leq \frac{k}{n} \right\}, \quad 1 \leq k \leq nN$$

$$e_n = \sum_{k=1}^{nN} \frac{k-1}{n} \mathbb{1}_{A_{k,n}} \text{ is measurable, } e_n(\omega) \leq f(\omega) < e_n(\omega) + \frac{1}{n}, \text{ and } f(\omega) = \lim_{n \rightarrow \infty} e_n(\omega).$$

// Finite linear combinations of indicators suffice to describe a measurable function.

2.2 Measure on a Measurable Set

Definition 2.1 (Additive and σ -Additive Function): Let Ω be a set, $\mathcal{A}$ a Boolean algebra over Ω . We say that $\mu : \mathcal{A} \rightarrow [0; \infty]$ is:

1. **additive**, if $\mu(\emptyset) = 0$ and $\mu(A \cup B) = \mu(A) + \mu(B)$ for $A \cap B = \emptyset$
2. **σ -additive**, if $\mu(\emptyset) = 0$ and $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$ for disjoint $\{A_i\}$ with $\bigcup A_i \in \mathcal{A}$

Definition 2.2 (Measure): A measure on a measurable space $(\Omega, \mathcal{F})$ is a σ -additive function $\mu : \mathcal{F} \rightarrow [0, \infty]$.

Remark 2.1: Let $\mu : \mathcal{F} \rightarrow [0, \infty]$ be a measure and $A_i \subseteq A_{i+1}$ for $i \in \mathbb{N}$, $A_i \in \mathcal{F}$. Then:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

If we set $B_1 = A_1$, $B_i = A_i \setminus A_{i-1}$, then $B_i \cap B_j = \emptyset$ for $i \neq j$, and:

$$\mu\left(\bigcup A_i\right) = \mu\left(\bigcup B_i\right) = \sum_{i=0}^{\infty} \mu(B_i) = \lim_{n \rightarrow \infty} \sum_{i=0}^n \mu(B_i) = \lim_{n \rightarrow \infty} \mu(A_n)$$

Remark 2.2: If $\mu(A_0) < \infty$ and $A_{i+1} \subseteq A_i$ and $\bigcap_{i=0}^{\infty} A_i = \emptyset$, then $\lim_{n \rightarrow \infty} \mu(A_n) = 0$.

$$(a, b] = (-\infty, b] \setminus (-\infty, a]; \quad (-\infty, \infty) = \bigcup_{n=1}^{\infty} (-n, n].$$

Remark 2.3: If $f, g : \Omega \rightarrow \mathbb{R}$ are measurable, then $h = \max(f, g)$ is measurable:

$$h^{-1}(-\infty, b] = \{\omega \mid f(\omega) \leq b\} \cap \{\omega \mid g(\omega) \leq b\} = f^{-1}(-\infty, b] \cap g^{-1}(-\infty, b]$$

$f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$ are measurable when f is measurable. $f^+, f^- \geq 0$, $f^+ \cdot f^- = 0$, $f = f^+ - f^-$, $|f| = f^+ + f^-$.

Definition 2.3 (Simple Function): Let $(\Omega, \mathcal{F})$ be measurable. We say that $\varphi : \Omega \rightarrow \mathbb{R}$ is **simple**, if $\varphi = \sum_{i=1}^n c_i \mathbb{1}_{A_i}$ for some $c_i > 0$ and disjoint $A_i \in \mathcal{F}$.

Definition 2.4 (Lebesgue Integral): Let $(\Omega, \mathcal{F})$ be measurable, $\mu : \mathcal{F} \rightarrow (0, \infty)$ a measure. We define the Lebesgue integral for $f : \Omega \rightarrow [0, \infty]$ measurable:

1. $f = \sum c_i \mathbb{1}_{A_i}$ is simple: $\int f d\mu = \sum c_i \mu(A_i)$ (well-defined, may be $+\infty$)
2. In the general case: $\int f d\mu = \sup \left\{ \int e d\mu \mid e \leq f, e \text{ is simple} \right\}$

2.3 Monotone Convergence Theorem

Theorem 2.4 (Monotone Convergence): Let $(f_n)_{n \in \mathbb{N}}$ be nonnegative measurable functions with $f_n \leq f_{n+1}$ for $\forall n$. Then:

1. $f : \Omega \rightarrow \mathbb{R}$, $f(\omega) = \lim_{n \rightarrow \infty} f_n(\omega)$ is measurable
2. $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$$

Proof. 1. $f_n(\omega) \leq f_{n+1}(\omega)$ for $\forall n \Rightarrow \lim f_n(\omega)$ exists.

$$\{\omega \mid f(\omega) \leq b\} = \{\omega \mid \lim_{n \rightarrow \infty} f_n(\omega) \leq b\} = \bigcap_{n \in \mathbb{N}} \{\omega \mid f_n(\omega) \leq b\} \in \mathcal{F}$$

2. If φ is simple and $\varphi \leq f_n$ (for some n), then $\int \varphi d\mu \leq \int f_n d\mu$. Since $f_n \leq f$: $\int \varphi d\mu \leq \int f d\mu$. $\Rightarrow \lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu$.

Conversely: if $\varphi \leq f$, φ is simple, we need to show $\int \varphi d\mu \leq \lim \int f_n d\mu$.

$\varphi = \sum_{i=1}^k c_i \mathbb{1}_{A_i}$. For $\varepsilon > 0$, $\varepsilon < c_i$:

$$B_m^{(\varepsilon)} = \{\omega \mid f_m(\omega) > \varphi(\omega) - \varepsilon\}$$

$$\bigcup B_m^{(\varepsilon)} = \Omega, B_m^{(\varepsilon)} \subseteq B_{m+1}^{(\varepsilon)}.$$

On $B_m^{(\varepsilon)}$: $f_m(\omega) \geq \sum_{i=1}^k (c_i - \varepsilon) \mathbb{1}_{A_i \cap B_m^{(\varepsilon)}}$, whence:

$$\int f_m d\mu \geq \sum_{i=1}^k (c_i - \varepsilon) \mu(A_i \cap B_m^{(\varepsilon)}) \xrightarrow{m \rightarrow \infty} \int \varphi d\mu$$

□

2.4 Corollary (Fatou's Theorem)

Theorem 2.5 (Fatou): Let $f_n \geq 0$ be measurable functions on $(\Omega, \mathcal{F})$ and μ a measure. Then:

$$\liminf_{n \rightarrow \infty} \int f_n d\mu \geq \int \liminf_{n \rightarrow \infty} f_n d\mu$$

Proof. Let $g_n = \inf_{m \geq n} f_m$. Then g_n is measurable; $g_n \geq 0$ and $f_n \geq g_n$. $g_n = \inf_{m \geq n} f_m \leq \inf_{m \geq n+1} f_m = g_{n+1} \Rightarrow g_n \uparrow$.

From the Monotone Convergence Theorem:

$$\lim_{n \rightarrow \infty} \int g_n d\mu = \int \lim_{n \rightarrow \infty} g_n d\mu = \int \liminf_{n \rightarrow \infty} f_n d\mu$$

Finally: $f_n \geq g_n \Rightarrow \int f_n d\mu \geq \int g_n d\mu$ and for $m \geq n$: $\inf_{m \geq n} \int f_m d\mu \geq \int g_n d\mu \rightarrow \int \liminf f_n d\mu$. □

Definition 2.5 (Integral of a Signed Function): Let $f : \Omega \rightarrow \mathbb{R}$ be measurable on $(\Omega, \mathcal{F})$ and $\mu : \mathcal{F} \rightarrow [0, \infty]$ a measure. If at least one of the two integrals $\int f^+ d\mu$ and $\int f^- d\mu$ is finite, then:

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

We say that f is **integrable**, if $\int |f| d\mu < \infty$.

Property 2.6: If f and g are measurable and integrable and $a \in \mathbb{R}$, then:

$$\int (af + g) d\mu = a \int f d\mu + \int g d\mu$$

2.5 Lebesgue's Theorem (Dominated Convergence)

Theorem 2.7 (Lebesgue): Let $(f_n)_{n \in \mathbb{N}}$ be measurable in $(\Omega, \mathcal{F})$, μ a measure and:

1. $f(\omega) = \lim_{n \rightarrow \infty} f_n(\omega)$ for $\omega \in \Omega$
2. g is integrable and for every n : $|f_n| \leq g$

Then $\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$.

Proof. Since $|f_n| \leq g$: $g - f_n \geq 0$ and $g + f_n \geq 0$. From Fatou's theorem:

$$\begin{aligned} \liminf \int (g - f_n) d\mu &\geq \int \liminf (g - f_n) d\mu \\ \int g d\mu + \liminf \int (-f_n) d\mu &\geq \int g d\mu - \int f d\mu \\ \Rightarrow \int f d\mu &\geq \limsup \int f_n d\mu \end{aligned}$$

Applying for $f = -f$, $f_n = -f_n$:

$$-\int f d\mu \geq -\liminf \int f_n d\mu \Rightarrow \limsup \int f_n d\mu \leq \int f d\mu \leq \liminf \int f_n d\mu$$

Hence everywhere $\Rightarrow \lim \int f_n d\mu = \int f d\mu$. □

Definition 2.6 (Probability Measure): Let $(\Omega, \mathcal{F})$ be measurable. We say that P is a **probability measure**, if:

1. P is a measure
2. $P(\Omega) = 1$

The expectation of a r.v. $X : \Omega \rightarrow \mathbb{R}$ with respect to P is $\mathbb{E}X \stackrel{\text{def}}{=} \int X dP$.

From last time: $\Phi_X(\lambda) = \log \mathbb{E} \exp(\lambda X)$, $X \in [a, b]$, $a, b \in \mathbb{R}$.

$$\frac{d}{d\lambda} \mathbb{E} \exp(\lambda X) = \frac{d}{d\lambda} \int \exp(\lambda X) dP = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int e^{(\lambda+h)X} dP - \int e^{\lambda X} dP \right)$$

Since $a \leq X \leq b$:

$$\lim_{n \rightarrow \infty} \mathbb{E} f_n = \mathbb{E} \lim f_n \quad \text{and hence:} \quad \frac{d}{d\lambda} \mathbb{E} e^{\lambda X} = \mathbb{E} \frac{d}{d\lambda} e^{\lambda X} = \mathbb{E} X e^{\lambda X}$$

(by Lebesgue's theorem).

3 Growth Function and Vapnik–Chervonenkis Dimension

$X, Y = \{0, 1\}$; Λ ; $h : \Lambda \times X \rightarrow \{0, 1\}$; $\ell : Y \times Y \rightarrow \{0, 1\}$, $\ell(y_1, y_2) = y_1 \oplus y_2$.

$Z = X \times Y = X \times \{0, 1\}$; $\bar{z} = (z_1, \dots, z_m) \in Z^m$, $z_i = (x_i, y_i)$.

For $\lambda \in \Lambda$:

$$S_\lambda(\bar{z}) = (S_\lambda(z_1), \dots, S_\lambda(z_m)) \in \{0, 1\}^m, \quad s_i = \ell(h(\lambda, x_i), y_i) = h(\lambda, x_i) \oplus y_i$$

Definition 3.1 (Growth Function): For $\bar{z} \in Z^m$: $S_\Lambda(\bar{z}) = \{S_\lambda(\bar{z}) \mid \lambda \in \Lambda\}$.

The **growth function** for Λ and h is $\Pi_\Lambda : \mathbb{N} \rightarrow \mathbb{N}$, defined by:

$$\Pi_\Lambda(m) = \max_{\bar{z} \in Z^m} |S_\Lambda(\bar{z})| \leq 2^m$$

Definition 3.2 (VCdim): The **Vapnik–Chervonenkis dimension** for Λ and h :

$$\text{VCdim}(\Lambda) = \max\{m \mid \Pi_\Lambda(m) = 2^m\}$$

- If $d = \text{VCdim}(\Lambda) < \infty$, then optimizing the empirical error gives a machine learning algorithm.
- If $d = \text{VCdim}(\Lambda) = \infty$, things are hopeless.

Remark 3.1: Intuition: VCdim measures the “expressibility” of the class Λ — how many points can be “shattered” in an arbitrary way. A large VCdim means a rich class (can overfit), a small one — a restricted class (better generalization). **Practically:** For a linear classifier in $\mathbb{R}^d$ we have $\text{VCdim} = d + 1$ — i.e. the number of parameters (weights) gives a direct measure of complexity. For neural networks VCdim grows as $O(|W| \log |W|)$, not polynomially in the number of layers.

See [Vap99]: ch. 3, pp. 80–89 (VC-dimension and growth function; [AB99]: ch. 3–4 (pp. 37–68); [MU05]: ch. (new ed.), section on VC-dimension.

Let $\bar{x} \in X^m$, $\tilde{S}_\lambda(\bar{x}) = (h(\lambda, x_1), \dots, h(\lambda, x_m))$. Then $|S_\Lambda(\bar{z})| = |\tilde{S}_\Lambda(\bar{x})|$.

Consider $X = \mathbb{R}^d$, $\Lambda = \mathbb{R}^d \times \mathbb{R}$:

$$h(\lambda, x) = \begin{cases} 1, & \langle (x, -1), \lambda \rangle \geq 0 \\ 0, & \text{otherwise} \end{cases} = \text{sgn}(\langle \lambda, (x, -1) \rangle), \quad \text{sgn}(a) = \begin{cases} 1, & a \geq 0 \\ 0, & a < 0 \end{cases}$$

Questions:

1. $\text{VCdim}(\Lambda) \leq d + 1$
2. $\Pi_\Lambda(m) = ?$
3. $\text{VCdim}(\Lambda) = d + 1$

3.1 Proposition: $\text{VCdim}(\Lambda) \leq d + 1$ (Theorem 3.1 — First Part)

Proposition 3.2: $\text{VCdim}(\Lambda) \leq d + 1$.

Proof. Since $|\tilde{S}_{\bar{x}}(\Lambda)| = |S_{\bar{x}}(\Lambda)|$, it suffices to consider only vectors $\bar{x} \in \mathbb{R}^d$.

Let $x_1, \dots, x_{d+2} \in \mathbb{R}^d$. We need $|S_{\bar{x}}(\Lambda)| < 2^{d+2}$.

Let $\hat{x}_i = (x_i, -1) \in \mathbb{R}^{d+1}$. Since there are $d + 2$ vectors in $\mathbb{R}^{d+1}$, they are linearly dependent: $\hat{x}_{d+2} = \sum_{i=1}^{d+1} \alpha_i \hat{x}_i$.

Suppose all vectors from $\{0, 1\}^{d+2}$ are elements of $\tilde{S}_{\bar{x}}(\Lambda)$.

For $\lambda \in \Lambda$:

$$\langle \hat{x}_{d+2}, \lambda \rangle = \left\langle \sum_{i=1}^{d+1} \alpha_i \hat{x}_i, \lambda \right\rangle = \sum_{i=1}^{d+1} \alpha_i \langle \hat{x}_i, \lambda \rangle$$

Then there exists $\lambda \in \mathbb{R}^{d+1}$: $\langle \hat{x}_i, \lambda \rangle = \text{sgn}(\alpha_i)$ for every $i \leq d + 1$. For such λ :

$$\langle \hat{x}_{d+2}, \lambda \rangle = \sum_{i=1}^{d+1} \alpha_i \text{sgn}(\alpha_i) = \sum |\alpha_i| \geq 0$$

$\Rightarrow \text{sgn}(\langle \hat{x}_{d+2}, \lambda \rangle) = 1 \Rightarrow (\text{sgn}(\alpha_1), \dots, \text{sgn}(\alpha_{d+1}), 0) \notin \tilde{S}_{\bar{x}}(\Lambda) \Rightarrow |\tilde{S}_{\bar{x}}(\Lambda)| < 2^{d+2} \Rightarrow \Pi_{\Lambda}(d + 2) < 2^{d+2} \Rightarrow \text{VCdim}(\Lambda) \leq d + 1$. $\square$

3.2 Connected Components and Formula for $|\tilde{S}_{\bar{x}}(\Lambda)|$

Let $x_1, \dots, x_m \in \mathbb{R}^d$. Define $\hat{x}_i = (x_i, -1) \in \mathbb{R}^{d+1}$.

$$P_i = \{\lambda \in \mathbb{R}^{d+1} \mid \langle \lambda, \hat{x}_i \rangle = 0\}$$

$\mathbb{R}^{d+1} \setminus \bigcup P_i$ splits into connected regions — connected components $CC(\mathbb{R}^{d+1} \setminus \bigcup P_i)$.

Definition 3.3 (Connectedness $\sim$): If $U \subseteq \mathbb{R}^{d+1}$, we say that x and y are connected in U if there exists a continuous function $f : [0, 1] \rightarrow U$ with $f(0) = x$ and $f(1) = y$. We write $x \sim y$. This is an equivalence relation and the equivalence classes of $\sim$ are **connected components**.

Proposition 3.3:

$$|S_{\bar{x}}(\Lambda)| = |CC(\mathbb{R}^{d+1} \setminus \bigcup P_i)|$$

Proof. 1. Let λ_1, λ_2 be in the same connected component. We claim $h(\lambda_1, x_i) = h(\lambda_2, x_i)$ for $\forall x_i$. From $\lambda_1 \sim \lambda_2$ there exists a continuous $g : [0, 1] \rightarrow \mathbb{R}^{d+1} \setminus \bigcup P_i$ with $g(0) = \lambda_1$, $g(1) = \lambda_2$. Then $\langle g(t), \hat{x}_i \rangle$ is continuous and does not pass through 0 $\Rightarrow$ it preserves sign by the intermediate value theorem.

$$F : CC(\mathbb{R}^{d+1} \setminus \bigcup P_i) \rightarrow \tilde{S}_{\bar{x}}(\Lambda), F([\lambda]_u) = (h(\lambda, x_1), \dots, h(\lambda, x_m)).$$

2. F is injective: Let $\lambda_1, \lambda_2 \in \mathbb{R}^{d+1} \setminus \bigcup P_i$ with $h(\lambda_1, x_i) = h(\lambda_2, x_i)$ for $\forall i$. Then $g(t) = (1 - t)\lambda_1 + t\lambda_2$ connects λ_1 with λ_2 .

3. F is surjective: If $v \in \tilde{S}_{\bar{x}}(\Lambda)$, there exists $\lambda \in \mathbb{R}^{d+1}$ with $v = (h(\lambda, x_1), \dots, h(\lambda, x_m))$. $\square$

3.3 Lemma (on the Number of Connected Components)

Lemma 3.4 (Lemma 3.3): Let $\hat{x}_1, \dots, \hat{x}_m \in \mathbb{R}^d$ be vectors in *general position* (i.e. every $r + 1 \leq d + 1$ of them have linear rank $r + 1$). Then:

$$|CC(\mathbb{R}^d \setminus \bigcup P_i)| = 2 \sum_{i=0}^{d-1} \binom{m-1}{i}$$

Proof. We use induction on m and d .

Base 1a: $d = 1$, $\hat{x}_i \neq 0 \Rightarrow P_i = \{0\}$. $\mathbb{R}^1 \setminus \{0\}$ has 2 connected regions. $2 \sum_{i=0}^0 \binom{0}{i} = 2 \cdot 1 = 2$. OK

Base 2a: $m = 1$, $\hat{x}_1 \neq 0$. $P_1 = \{\lambda \mid \langle \lambda, \hat{x}_1 \rangle = 0\}$ — a hyperplane. $H^+ = \{\lambda \mid \langle \lambda, \hat{x}_1 \rangle > 0\}$, $H^- = \{\lambda \mid \langle \lambda, \hat{x}_1 \rangle < 0\}$. $|CC(\mathbb{R}^{d+1} \setminus P_1)| = 2 = 2 \sum_{i=0}^0 \binom{0}{i}$. OK

Induction step: Let $C_1, \dots, C_N$ be the components of $\mathbb{R}^{d+1} \setminus \bigcup_{i=1}^m P_i$. $N' = |CC(\mathbb{R}^{d+1} \setminus \bigcup_{j=1}^{m+1} P_j)|$.

Let:

$$I^+ = \{i \mid C_i \cap P_{m+1} \neq \emptyset\}, \quad I^- = \{i \mid C_i \cap P_{m+1} = \emptyset\}$$

1. $N' = |I^-| + 2|I^+|$
2. $|I^-| + |I^+| = |CC(\mathbb{R}^{d+1} \setminus \bigcup_{i=1}^m P_i)| = N$
3. $|I^+| = |CC(P_{m+1} \setminus \bigcup_{i=1}^m P_i)|$

Then:

$$\begin{aligned} N' &= |I^-| + 2|I^+| = (|I^-| + |I^+|) + |I^+| = N + |CC(P_{m+1} \setminus \bigcup P_i)| \\ &\stackrel{\text{i.h.}}{=} 2 \sum_{i=0}^{d-1} \binom{m-1}{i} + 2 \sum_{i=0}^{d-2} \binom{m-1}{i} = 2 \sum_{i=0}^d \binom{m-1}{i} \end{aligned}$$

by Pascal's formula $\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}$. □

3.4 Theorem for $|\tilde{S}_{\bar{x}}(\Lambda)|$ and Sauer's Lemma

Proposition 3.5 (Theorem 3.1): Let $x_1, \dots, x_m$ be points in $\mathbb{R}^d$ in general position. Then $|\tilde{S}_{\bar{x}}(\Lambda)| = 2 \sum_{i=0}^d \binom{m-1}{i}$.

Proposition 3.6: In the general case: $|\tilde{S}_{\bar{x}}(\Lambda)| \leq 2 \sum_{i=0}^d \binom{m-1}{i}$ (by induction).

Corollary 3.7: $|\tilde{S}_{\bar{x}}(\Lambda)| = 2^m$ for $m \leq d + 1$ and $\bar{x}$ in general position. For $m > d + 1$: $|\tilde{S}_{\bar{x}}(\Lambda)| \leq 2 \sum_{i=0}^d \binom{m-1}{i} < 2^m$.

Therefore:

$$\Pi_{\Lambda}(m) = \begin{cases} 2^m, & m \leq d + 1 \\ < 2^m, & m > d + 1 \end{cases} \Rightarrow \text{VCdim}(\Lambda) = d + 1$$

Lemma 3.8 (Sauer, Theorem 3.6): Let $\text{VCdim}(\Lambda) = d$. Then for every $\bar{z} \in Z^m$:

$$|S_{\bar{z}}(\Lambda)| \leq \sum_{i=0}^d \binom{m}{i}$$

Better bound: $\sum_{i=0}^d \binom{m}{i} \leq \left(\frac{me}{d}\right)^d$ for $m \geq d$.

Remark 3.9: Significance of Sauer's Lemma: It gives a polynomial upper bound $\Pi_{\Lambda}(m) \leq (me/d)^d$ instead of the exponential 2^m — a “phase transition” at $m = d$. This is the key statement that makes VCdim practically useful: for $m \gg d$ the class makes far fewer “distinct predictions” on m points than is theoretically possible.

See [Vap99]: ch. 3, pp. 77–80 (Theorem 3.1, Sauer–Shelah); [AB99]: ch. 3, pp. 46–52.

Proof of Sauer's Lemma. Let $S \subseteq \{0, 1\}^m$, $d(S) = \max\{|I| \mid I \subseteq \{1, \dots, m\}, |S_I| = 2^{|I|}\}$. By assumption $d(S) \leq \text{VCdim}(\Lambda) = d$.

By induction on $m + d(S)$:

Base:

- $d' = 0$: $|S| \leq 1 = \binom{m}{0}$.
- $m = 1$: $S \subseteq \{0, 1\} \Rightarrow |S| \leq 2 = \binom{1}{0} + \binom{1}{1}$.

I.H.: $S^+ = \{s \in \{0, 1\}^m \mid (s, 0), (s, 1) \in S\}$, $S^- = \{s \in \{0, 1\}^m \mid (s, 0) \in S, (s, 1) \notin S\}$.

$$|S| = |S^+| + |S^-|, \quad d(S^-) \leq d(S), \quad d(S^+) \leq d(S) - 1$$

$$|S| = |S^+| + |S^-| \leq \sum_{i=0}^{d-1} \binom{m-1}{i} + \sum_{i=0}^d \binom{m-1}{i} = \sum_{i=0}^d \binom{m}{i}$$

□

4 Existence of a Learning Algorithm with Finite VCdim

Under assumption (*):

$$\hat{\rho}(\lambda, \bar{z}) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(\lambda, x_i) \neq y_i}, \quad \rho(\lambda) = \mathbb{E}[\mathbb{1}_{h(\lambda, x) \neq y}]$$

$$\hat{\lambda} := \hat{\lambda}(\bar{z}) = \operatorname{argmin}_{\lambda \in \Lambda} \hat{\rho}(\lambda, \bar{z}), \quad \lambda^* := \operatorname{argmin}_{\lambda \in \Lambda} \rho(\lambda)$$

It suffices to control $P(\sup_{\lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \varepsilon) \leq \delta$.

Steps:

1. **Symmetrization:** adding an independent copy $\bar{z}' \sim \bar{z}$:

$$P\left(\sup_{\lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \varepsilon\right) \leq 2 P\left(\sup_{\lambda} |\hat{\rho}(\lambda, \bar{z}') - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{2}\right)$$

2. **Averaging + uniform distribution over permutations of $(\bar{z}, \bar{z}')$**
3. **Replacing P (unknown) with something known**
4. **Reduction to the finite case:** The differences $\hat{\rho}(\lambda, \bar{z}) - \hat{\rho}(\lambda, \bar{z}')$ are determined by $\tilde{S}(\Lambda)$.

Proposition 4.1: Let Λ be countable. Then for every $\varepsilon > 0$ and $m > \frac{2}{\varepsilon^2}$:

$$P^m\left(\sup_{\lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| > \varepsilon\right) \leq 2 P^{2m}\left(\sup_{\lambda} |\hat{\rho}(\lambda, \bar{z}') - \hat{\rho}(\lambda, \bar{z})| > \frac{\varepsilon}{2}\right)$$

4.1 Rademacher Lemma and Corollary

Lemma 4.2: Let $\bar{z} \in Z^{2m}$. Then:

$$\Pr(\sigma \bar{z} \in S) \leq 2 \Pi_{\Lambda}(2m) \exp\left(-\frac{m\varepsilon^2}{8}\right)$$

Corollary 4.3: $\text{VCdim}(\Lambda) = d$, $\varepsilon > 0$, $m > \frac{2}{\varepsilon^2}$, $2m > d$. Then:

$$P\left(\sup_{\lambda} |\rho(\lambda) - \hat{\rho}(\lambda, \bar{z})| \geq \varepsilon\right) \leq 4 \exp\left(-\frac{m\varepsilon^2}{8} + d \ln\left(\frac{2m}{d} + 1\right)\right)$$

Remark 4.4: Practical significance: This bound gives

$$P(\text{error}) \leq 4 \exp\left(-\frac{m\varepsilon^2}{8} + d \ln \frac{2m}{d}\right)$$

i.e. for $m \gg d \ln(m/d)/\varepsilon^2$ the probability of poor generalization is small.

Note that the required sample size m grows only *logarithmically* with m in the right-hand side, so $m = O(d \ln(1/\varepsilon)/\varepsilon^2)$ examples suffice.

See [Vap99]: ch. 3, pp. 83–90; [AB99]: ch. 5 (PAC-bounds), pp. 69–95.

4.2 General Bounds for Finite VCdim

Let $\varepsilon > 0$, $\delta > 0$, $d = \text{VCdim}(\Lambda)$.

$$P\left(\hat{\rho}(\hat{\lambda}) - \rho(\lambda^*) > \varepsilon\right) \leq 8 \exp\left(-\frac{m\varepsilon^2}{32} + d(\ln(2m) + 1)\right)$$

1. $m_0(\varepsilon, \delta)$: If $m > m_0(\varepsilon, \delta)$, then $P(|\hat{\rho}(\hat{\lambda}) - \rho(\lambda^*)| > \varepsilon) \leq \delta$. It suffices that:

$$8 \exp\left(-\frac{m\varepsilon^2}{32} + d(\ln(2m) + 1)\right) \leq \delta$$

At $m = \frac{64d}{\varepsilon^2} \ln \frac{8}{\delta}$ it is sufficient.

2. $\varepsilon_0(m, \delta)$: Given m , δ :

$$\varepsilon_0 = \sqrt{\frac{32(d(\ln(2m) + 1) + \ln(8/\delta))}{m}}$$

4.3 Generalization for $\ell : X \times Y \rightarrow [0, B]$

$B \in \mathbb{R}^+$, $\ell : X \times Y \rightarrow [0, B]$; $e(\lambda, z_i) = \ell(h(\lambda, x_i), y_i)$.

For fixed $N \in \mathbb{N}$: $\varphi_N(\lambda, z_i) = \lfloor Ne(\lambda, z_i) \rfloor / N \uparrow$ and $\mathbb{E}\varphi_N \rightarrow \mathbb{E}e$.

We obtain:

$$\frac{1}{m} \sum e(\lambda, z_i) - \mathbb{E}e(\lambda, z) \leq B \sup_{p \in [0, B]} \left| \frac{1}{m} \sum \mathbb{1}_{e(\lambda, z_i) > p} - \mathbb{E} \mathbb{1}_{e(\lambda, z) > p} \right|$$

$$P\left(\sup_{\lambda} |\hat{f}(\lambda, \bar{z}) - f(\lambda)| \geq \varepsilon\right) \leq 4 \Pi_{\Lambda}(2m) \exp\left(-\frac{\varepsilon^2 m}{8B^2}\right) \leq 4 \exp\left(-\frac{\varepsilon^2 m}{8B^2} + d\left(\ln \frac{2m}{d} + 1\right)\right)$$

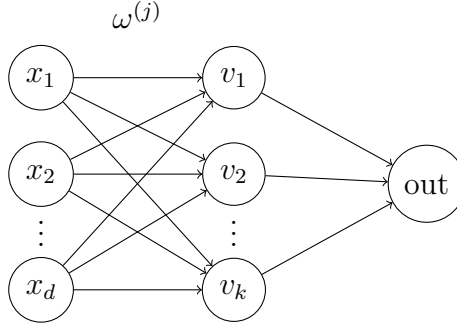
5 Acyclic Neural Network

$$\text{sgn}(\langle x, \omega \rangle - \theta)$$

Acyclic neural network (classifier) with input $x \in \mathbb{R}^d$, vertices $v_1, \dots, v_k$ (hidden layer), weights ω . The value at v_ℓ is computed as:

$$\text{sgn}(\langle y^{(j)}, \omega^{(j)} \rangle - \theta^{(j)})$$

where $y^{(j)}$ are the values of the vertices with an edge going into v_ℓ ; $\omega^{(j)}$ — the weights of those edges; $\theta^{(j)}$ — the bias.



We can assume that $v_1, \dots, v_k$ are ordered topologically. Δ_i — number of edges entering v_i ; $S_p(i)$ — number of distinct functions (vectors) of the form $(v_i(1), \dots, v_i(m))$.

$$S_x(i) \leq \left(\frac{me}{\Delta_i}\right)^{\Delta_i}, \quad \Pi S_x(i) \leq \prod_{i=1}^k \left(\frac{me}{\Delta_i}\right)^{\Delta_i+1}$$

$|W| = \sum(\Delta_i + 1)$ and:

$$\ln \prod \left(\frac{me}{\Delta_i}\right)^{\Delta_i+1} = \sum(\Delta_i + 1) \ln \frac{me}{\Delta_i} \leq |W| \ln \frac{eme}{|W|}$$

(by Jensen's inequality).

Therefore:

$$S_x(W) \leq \left(\frac{eme}{|W|}\right)^{|W|}$$

— for our neural network, which asymptotically does not differ from a single-layer neural network (perceptron).

6 Product of Measurable Spaces and Measures

Definition 6.1 ($\mathcal{F}_1 \otimes \mathcal{F}_2$): Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable. Then with $\Omega = \Omega_1 \times \Omega_2$:

$$\mathcal{F} = \mathcal{F}_1 \otimes \mathcal{F}_2 = \sigma(A \times B \mid A \in \mathcal{F}_1, B \in \mathcal{F}_2)$$

(the smallest sigma-algebra containing the rectangles).

$\mathcal{A} = \{\bigcup (A_i \times B_i) \mid n \in \mathbb{N}, A_i \in \mathcal{F}_1, B_i \in \mathcal{F}_2, (A_i \times B_i) \cap (A_j \times B_j) = \emptyset (i \neq j)\}$
generates $\mathcal{F}_1 \otimes \mathcal{F}_2$. $\mathcal{A}$ is an algebra, because:

1. $\Omega \setminus (A \times B) = \bar{A} \times B \cup A \times \bar{B} \cup \bar{A} \times \bar{B} \in \mathcal{A}$
2. $(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2) \in \mathcal{A}$
3. Intersection of sets from $\mathcal{A}$ is in $\mathcal{A}$
4. $\mathcal{A}$ is closed under complementation

Definition 6.2 (Product of Measures): $\mu(\bigcup (A_i \times B_i)) = \sum \mu_1(A_i) \mu_2(B_i)$.

Theorem 6.1 (Carathéodory): If $\mathcal{A}$ is a Boolean algebra and μ is σ -additive, $\mu : \mathcal{A} \rightarrow [0, \infty]$, then μ can be extended to a measure on $\sigma(\mathcal{A})$. If $\mu(\Omega) < \infty$ or μ is σ -finite, the extension is unique.

Exercise: If $A \times B = \bigcup (A_i \times B_i)$ (disjoint), prove that $\mu_1(A) \mu_2(B) = \sum \mu_1(A_i) \mu_2(B_i)$.

Idea: $f(a) = \sum \mu_2(B_i) \mathbb{1}_{A_i}(a) = \mathbb{1}_{A \times B} \cdot \mu_2(B)$. On one side: $\int f d\mu_1 = \mu_1(A) \mu_2(B)$. On the other: $\int f d\mu_1 = \sum \mu_2(B_i) \mu_1(A_i)$.

6.1 Measure Zero and “Almost Everywhere”

We say that A has measure zero if $\mu(A) = 0$. We say that a property of $\omega \in \Omega$ holds **almost everywhere**, if the set $\{\omega \mid \text{the property does not hold}\}$ has measure zero.

Theorem 6.2 (Tonelli): Let $f : \Omega_1 \times \Omega_2 \rightarrow [0, \infty]$ be measurable in $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, \mu)$, $\mu = \mu_1 \otimes \mu_2$, μ_1, μ_2 — finite or σ -finite. For $\omega_1 \in \Omega_1$, $f_{\omega_1}(\omega_2) = f(\omega_1, \omega_2)$. Then:

1. For $\forall \omega_1 \in \Omega_1$, f_{ω_1} is measurable
2. $F(\omega_1) = \int f_{\omega_1} d\mu_2$ is $(\Omega_1, \mathcal{F}_1, \mu_1)$ -measurable
3. $\int_{\Omega_1 \times \Omega_2} f d\mu = \int_{\Omega_1} F d\mu_1$

Theorem 6.3 (Fubini): Let $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ be integrable in $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, \mu)$, $\mu = \mu_1 \otimes \mu_2$. Then:

1. For $\forall \omega_1$ — f_{ω_1} is integrable in $(\Omega_2, \mathcal{F}_2, \mu_2)$ (outside a set of measure zero)

2. $F(\omega_1) = \int f_{\omega_2} d\mu_2$ is integrable in $(\Omega_2, \mathcal{F}_2, \mu_2)$

3. $\int_{\Omega_2} F d\mu_2 = \int_{\Omega_1 \times \Omega_2} f d\mu$

Example: $\mathbb{E}X$, $X \geq 0$ with distribution P :

$$\mathbb{E}X = \int X(\omega) dP = \int \int_0^\infty \mathbb{1}_{u \leq X} du dP \stackrel{\text{Tonelli}}{=} \int \int \mathbb{1}_{u \leq X} dP du = \int_0^\infty P(X \geq u) du$$

More generally (f differentiable, $X \geq 0$):

$$\mathbb{E}f(X) = \int_0^\infty f'(u)P(X \geq u) du$$

6.2 L^p Spaces

Definition 6.3 ($\|\cdot\|_p, L^p$): Let $(\Omega, \mathcal{F}, \mu)$, μ σ -finite, $p \in [1, \infty)$. For a measurable function f we define:

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p}$$

$L^p = \{f \text{ measurable} \mid \|f\|_p < \infty\}$.

Theorem 6.4 (Hölder): 1. **Young:** If $a, b > 0$ and $p, q > 1$: $\frac{1}{p} + \frac{1}{q} = 1$, then:

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab$$

2. **Hölder:** $\frac{1}{p} + \frac{1}{q} = 1$ for $p, q > 1$. Then:

$$\int |fg| d\mu \leq \|f\|_p \|g\|_q$$

Proof. 1. $\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \ln(a^p) + \frac{1}{q} \ln(b^q) = \ln a + \ln b = \ln ab$ (by concavity).

2. Let $\tilde{f} = |f|/\|f\|_p$, $\tilde{g} = |g|/\|g\|_q$, $\|\tilde{f}\|_p = 1$, $\|\tilde{g}\|_q = 1$.

$$\int \tilde{f} \tilde{g} d\mu \leq \int \left(\frac{\tilde{f}^p}{p} + \frac{\tilde{g}^q}{q} \right) d\mu = \frac{1}{p} \int \tilde{f}^p d\mu + \frac{1}{q} \int \tilde{g}^q d\mu = \frac{1}{p} + \frac{1}{q} = 1$$

$$\Rightarrow \int \frac{|f| \cdot |g|}{\|f\|_p \|g\|_q} d\mu \leq 1 \Rightarrow \int |fg| d\mu \leq \|f\|_p \|g\|_q$$

□

Theorem 6.5 (Minkowski — triangle inequality): Let $p \in [1, \infty)$, $\|f\|_p, \|g\|_p < \infty$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Proof.

$$\|f + g\|_p^p = \int |f + g|^p d\mu \leq \int |f| \cdot |f + g|^{p-1} d\mu + \int |g| \cdot |f + g|^{p-1} d\mu$$

By Hölder (with $q = p/(p-1)$):

$$\begin{aligned} &\leq \|f\|_p \cdot \| |f + g|^{p-1} \|_q + \|g\|_p \cdot \| |f + g|^{p-1} \|_q = (\|f\|_p + \|g\|_p) \cdot \|f + g\|_p^{p-1} \\ &\Rightarrow \|f + g\|_p \leq \|f\|_p + \|g\|_p \end{aligned}$$

□

Definition 6.4 (Linear Space and Norm): A linear space

$(V, +, \cdot, 0)$ over $\mathbb{R}$: $u, v \in V \Rightarrow u + v \in V$ and $cu \in V$ ($c \in \mathbb{R}$), and the standard axioms.

A norm on V : $\|\cdot\| : V \rightarrow [0, \infty)$ is:

1. $\|u + v\| \leq \|u\| + \|v\|$
2. $\|cu\| = |c| \cdot \|u\|$
3. $\|u\| = 0 \Leftrightarrow u = 0$

$\|\cdot\|$ is a **seminorm** if (1) and (2) and $\|0\| = 0$.

Example 6.6: $I = [0, 1]^d$, $C(I) = \{f : I \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, $\|f\|_\infty = \max_{x \in I} |f(x)|$.

Definition 6.5 (Banach Space): A Cauchy sequence in $(V, \|\cdot\|)$ is a sequence (v_n) for which $\forall \varepsilon > 0 \exists n : \|v_{m_1} - v_{m_2}\| < \varepsilon$ for $m_1, m_2 \geq n$.

$(V, \|\cdot\|)$ is **Banach** if every Cauchy sequence has a limit.

Theorem 6.7 (Riesz–Fisher): Let $(\Omega, \mathcal{F}, \mu)$ be measurable, $p \in [1, \infty)$, (f_n) a Cauchy sequence in $\|\cdot\|_p$. Then there exists f measurable: $\|f\|_p < \infty$ and $\|f_n - f\|_p \rightarrow 0$.

6.3 Hilbert Space and Conditional Expectation

Definition 6.6 (Inner Product Space): $(V, \langle \cdot, \cdot \rangle)$ is an inner product space if $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfies:

1. $\langle u, v \rangle = \langle v, u \rangle$
2. $\langle au + v, w \rangle = a\langle u, w \rangle + \langle v, w \rangle$
3. $\langle u, u \rangle \geq 0$ with “=” $\Leftrightarrow u = 0$

The norm: $\|v\| = \sqrt{\langle v, v \rangle}$.

Definition 6.7 (Hilbert Space): $(V, \langle \cdot, \cdot \rangle)$ is **Hilbert**, if $(V, \|\cdot\|)$ is Banach.

Proposition 6.8: Let $(V, \langle \cdot, \cdot \rangle)$ be a Hilbert space, and $C \subseteq V$ a closed convex set ($c_1, c_2 \in C$, $d \in (0, 1) \Rightarrow dc_1 + (1 - d)c_2 \in C$). Then $\Pi_C : V \rightarrow C$, $\Pi_C(v) = \operatorname{argmin}_{u \in C} \|u - v\|$, is well-defined. If additionally C is linear, then $\forall c \in C$: $\langle v - \Pi_C(v), c - \Pi_C(v) \rangle = 0$.

Definition 6.8 (Linear Functional): If $(V, \|\cdot\|)$ is Banach, a linear functional is $L : V \rightarrow \mathbb{R}$ with $L(av + u) = aL(v) + L(u)$. $\|L\| = \sup \|L(v)\|/\|v\|$.

Theorem 6.9 (Riesz–Fréchet): If $(V, \langle \cdot, \cdot \rangle)$ is Hilbert and $L : V \rightarrow \mathbb{R}$ is linear with $\|L\| < \infty$, then there exists $g \in V$: $\forall f \in V$ ($L(f) = \langle f, g \rangle$).

See [AB99]: Appendix A (functional analysis), pp. 295–310; standard textbooks on functional analysis (Rudin, Yosida).

Theorem 6.10 (Radon–Nikodym): Let $(\Omega, \mathcal{F})$ be measurable and μ, ν finite (σ -finite) measures with $\forall A \in \mathcal{F}$ ($\mu(A) = 0 \Rightarrow \nu(A) = 0$) (ν is “absolutely continuous”). Then there exists a measurable μ -integrable function f : $\forall A \in \mathcal{F}$ $\int_A f d\mu = \nu(A)$.

See [AB99]: Appendix A; standard textbooks on measure theory (Halmos, Royden).

6.4 Conditional Expectation

Definition 6.9 (Conditional Expectation): Let $Z : \Omega \rightarrow \mathbb{R}$ be a random variable on $(\Omega, \mathcal{F}, P)$. Let $G \subseteq \mathcal{F}$. We say that Y is the **conditional expectation** of $Z|G$, if:

- Y is G -measurable
- For every $A \in G$: $\int_A Y dP = \int_A Z dP$

We write $Y = \mathbb{E}[Z | G]$.

Proposition 6.11: If $Z : \Omega \rightarrow \mathbb{R}$ is integrable and $G \subseteq \mathcal{F}$ is a σ -algebra, then:

1. There exists Y , a G -measurable function
2. If Y_1 and Y_2 both have this property, then $P(Y_1 \neq Y_2) = 0$

Property 6.12 (Tower property): If $D \subseteq G \subseteq \mathcal{F}$ are σ -algebras and Z is integrable, then:

$$\mathbb{E}[\mathbb{E}[Z | G] | D] = \mathbb{E}[Z | D]$$

Property 6.13: If $X_n \uparrow X$, $X_n \geq 0$ and X is integrable, then for $G \subseteq \mathcal{F}$:

$$\mathbb{E}[X | G] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n | G]$$

Property 6.14 (The known pulls out):

$$\mathbb{E}[XY | G] = Y \cdot \mathbb{E}[X | G], \quad \text{when } Y \text{ } G\text{-measurable}$$

6.5 Independence

Definition 6.10 (Independent Events): $(\Omega, \mathcal{F}, P)$ is a probability space. We say that events $A_1, \dots, A_n \in \mathcal{F}$ are **independent** (jointly) if for every $\emptyset \subseteq I \subseteq \{1, \dots, n\}$:

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Definition 6.11 (Independent Random Variables): $X_1, \dots, X_n$ are **independent**, if for every $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$, the events $X_i^{-1}(A_i)$ are independent.

If X, Y are independent:

$$P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B) = P_{X,Y}(A \times B)$$

$$\sigma(X, Y) = \sigma(X) \otimes \sigma(Y).$$

In particular: $\mathbb{E}f(X, Y) = \iint f(x, y) dP_X dP_Y$.

Remark 6.15: If X, Y are independent, f, g are measurable: $\mathbb{E}(f(X)g(Y)) = \mathbb{E}f(X) \cdot \mathbb{E}g(Y)$.

7 Rademacher Complexity

$h : \Lambda \times X \rightarrow Y; \ell : Y \times Y \rightarrow [0, B]; \bar{z} = (z_1, \dots, z_m);$

$$\hat{f}(\lambda, \bar{z}) = \frac{1}{m} \sum_{i=1}^m \ell(h(\lambda, x_i), y_i), \quad f(\lambda) = \mathbb{E} \ell(h(\lambda, x), y)$$

Goal: $P\left(\sup_{\lambda} |f(\lambda) - \hat{f}(\lambda, \bar{z})| > \varepsilon\right).$

Steps:

1. Showing that sup is not too far from $\mathbb{E} \sup$
2. $\mathbb{E} \sup_{\lambda} [\hat{f}(\lambda, z) - \mathbb{E} \hat{f}(\lambda, z')] \rightarrow$ symmetrization
3. Adding $\varepsilon_i \sim \text{Be}(\frac{1}{2})$, $\varepsilon_i \in \{-1, 1\}$ — the Rademacher random variables:

$$\mathbb{E}_z \mathbb{E}_{z'} \sup_{\lambda} \frac{1}{m} \sum \varepsilon_i \ell(h(\lambda, x_i), y_i) =: R_m(\Lambda, \ell)$$

Theorem 7.1 (McDiarmid's Inequality): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $c_1, \dots, c_n > 0$ with:

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i$$

If $X_1, \dots, X_n$ are independent random variables, then for $\varepsilon > 0$:

$$P(f(X_1, \dots, X_n) - \mathbb{E} f > \varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum c_i^2}\right)$$

In the special case $f(x_1, \dots, x_n) = \sum x_i$ we recover exactly Hoeffding.

$$|f(\bar{z}) - f(z_1, \dots, z'_i, \dots, z_m)| \leq \frac{B}{m}$$

Hence:

$$P\left(\sup_{\lambda} G(\lambda, z) - \mathbb{E} \sup_{\lambda} G(\lambda, z) \geq \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2 m}{B^2}\right)$$

where $G(\lambda, z) = \hat{f}(\lambda, z) - f(\lambda)$.

7.1 Definition of Rademacher Complexity

Definition 7.1 ($R_m(\mathcal{F}), \hat{R}_m(\mathcal{F})$ - Rademacher Complexity): Let $\mathcal{F} \subset \{f : Z \rightarrow \mathbb{R}\}$. We define:

$$R_m(\mathcal{F}) = \mathbb{E}_{\bar{z}} \mathbb{E}_{\varepsilon \in \{-1, 1\}^m} \sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^m \varepsilon_i f(z_i)$$

Empirical Rademacher Complexity:

$$\hat{R}_m(\mathcal{F}) = \mathbb{E}_{\varepsilon \in \{-1,1\}^m} \sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^m \varepsilon_i f(z_i) \mid \bar{z}$$

Theorem 7.2 (T05): With probability $\geq 1 - \delta$:

$$\hat{\rho}(\hat{\lambda}) - \rho(\lambda^*) \leq 4R_m(\mathcal{F}) + B\sqrt{\frac{2\ln(2/\delta)}{m}}$$

Remark 7.3: Advantage of Rademacher complexity over VCdim: The bound via $R_m(\mathcal{F})$ is *data-dependent* — it depends on the specific data $\bar{z}$, not just on the class $\mathcal{F}$. This makes it sharper in practice: if the data are “easy”, R_m is small even when VCdim is large. **Theoretically:** $R_m(\mathcal{F}) \leq B\sqrt{2d\ln(2m)/m}$ when VCdim = d , i.e. it is at least as good as the VC-bound.

See [AB99]: ch. 28–29 (pp. 209–240); Bartlett & Mendelson, JMLR 2002 (Rademacher generalization); [MU05]: new ed., section on Rademacher averages.

Property 7.4 (Properties of Rademacher Complexity): 1. $R_m(c\mathcal{F}) = |c|R_m(\mathcal{F})$ for $c \in \mathbb{R}$

2. $R_m(\mathcal{F}_1 + \mathcal{F}_2) \leq R_m(\mathcal{F}_1) + R_m(\mathcal{F}_2)$

Proposition 7.5 (Property for Linear Functions): Let $Z \in \mathbb{R}^d$, $\|z\|_2 \leq a$, $\mathcal{F} = \{f : Z \rightarrow \mathbb{R} \mid \exists \lambda \in B(1), f(z) = \langle \lambda, z \rangle\}$. Then:

$$\hat{R}_m(\mathcal{F}) \leq \frac{a}{\sqrt{m}}, \quad R_m(\mathcal{F}) \leq \frac{a}{\sqrt{m}}$$

Proof.

$$\begin{aligned} \hat{R}_m(\mathcal{F}) &= \mathbb{E} \sup_{\|\lambda\| \leq 1} \frac{1}{m} \sum_{i=1}^m \varepsilon_i \langle \lambda, z_i \rangle = \mathbb{E} \left\| \frac{1}{m} \sum \varepsilon_i z_i \right\|_2 \leq \frac{1}{m} \sqrt{\mathbb{E} \left\| \sum \varepsilon_i z_i \right\|_2^2} \\ &= \frac{1}{m} \sqrt{\sum_i \|z_i\|^2} = \frac{1}{m} \sqrt{m \cdot a^2} = \frac{a}{\sqrt{m}} \end{aligned}$$

□

7.2 McDiarmid

Lemma 7.6 (McDiarmid): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $c_1, \dots, c_n \in [0, \infty)$: for every $i \leq n$ and $x, x' \in \mathbb{R}^n$: $|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i$. If $X_1, \dots, X_n$ are independent random variables and $\varepsilon > 0$:

$$P(f(X) - \mathbb{E}f(X) \geq \varepsilon) \leq \exp\left(-\frac{2\varepsilon^2}{\sum c_i^2}\right)$$

Proof. 1. $f(X)$ is bounded (since $|f(y) - f(x)| \leq \sum c_i$).

2. Let $Y_i := \mathbb{E}[f(X) \mid X_1, \dots, X_i]$. Then $Y_n = f(X)$, $Y_0 = \mathbb{E}f(X)$.

$$f(X) - \mathbb{E}f(X) = Y_n - Y_0 = \sum_{i=1}^n (Y_i - Y_{i-1})$$

3. Goal: $P(\sum (Y_i - Y_{i-1}) \geq \varepsilon) \leq \frac{\mathbb{E} \exp(\lambda \sum (Y_i - Y_{i-1}))}{\exp(\lambda \varepsilon)}$.

4. $\mathbb{E} \exp(\lambda(Y_n - Y_0)) = \mathbb{E} \prod \exp(\lambda(Y_i - Y_{i-1}))$.

$Y_{i-1} = \mathbb{E}[Y_i \mid X_1, \dots, X_{i-1}]$ and for fixed $X_1, \dots, X_{i-1}$: $Y_i(X_1, \dots, X_{i-1}, X_i)$ takes values in an interval of length $\leq c_i$.

By Hoeffding: $\log \mathbb{E} \exp(\lambda(Y_i - Y_{i-1})) \leq \frac{\lambda^2 c_i^2}{8}$.

5. By the tower property:

$$\mathbb{E} \prod \exp(\lambda(Y_i - Y_{i-1})) \leq \prod_{i=1}^n \exp\left(\frac{\lambda^2 c_i^2}{8}\right) = \exp\left(\frac{\lambda^2 \sum c_i^2}{8}\right)$$

$$P(f(X) - \mathbb{E}f(X) \geq \varepsilon) \leq \frac{\exp(\lambda^2 \sum c_i^2 / 8)}{\exp(\lambda \varepsilon)}$$

At $\lambda^* = \frac{4\varepsilon}{\sum c_i^2}$:

$$P \leq \exp\left(-\frac{2\varepsilon^2}{\sum c_i^2}\right)$$

□

Remark 7.7: Intuition for McDiarmid: The inequality generalizes Hoeffding by replacing “independent differences” with “a function with bounded influence of each coordinate”. The number c_i measures how much a single example z_i can change the output — if c_i is small, the function is “stable” and cannot deviate much from its mean. **Application in ML:** For $f = \sup_{\lambda} |\hat{\rho} - \rho|$ we have $c_i = B/m$ (changing one example changes the average by at most B/m) — from which the bound $\exp(-2\varepsilon^2 m / B^2)$ follows.

See [AB99]: ch. 8, pp. 129–145 (McDiarmid and Rademacher); [MU05]: ch. 13 (martingales, pp. 298–320), Theorem 13.14 (Azuma–Hoeffding, equivalent formulation).

7.3 Summary

Scientific approaches for estimating the difference between the empirical and true error:

- VCdim $\rightarrow d$, δ : $\ell : X \times Y \rightarrow [0, B]$
- Rademacher $\rightarrow$ Lipschitzness, $\|z\| \leq a$, $\|\lambda\| \leq 1$: $\ell : X \times Y \rightarrow [0, B]$

8 Expressibility of Models

→ Cybenko: Every continuous function $f : [0, 1]^d \rightarrow \mathbb{R}$ can be approximated by a linear combination of sigmoids.

- Functional analysis — Hahn–Banach, Riesz theorems
- Signed measure
- Fourier transform, Fourier series

8.1 Hahn–Banach

Theorem 8.1 (Hahn–Banach): Let $(V, \|\cdot\|)$ be a Banach space and $p : V \rightarrow [0, \infty)$ with: $p(\lambda v) = \lambda p(v)$ for $\lambda \geq 0$; $p(u + v) \leq p(u) + p(v)$.

Let $U \subseteq V$ be a closed linear subspace and $L : U \rightarrow \mathbb{R}$ a linear functional with $L(u) \leq p(u)$ for $\forall u \in U$.

Then L can be extended to a linear $L' : V \rightarrow \mathbb{R}$ such that:

1. $L'(u) = L(u)$ for $u \in U$
2. $L'(v) \leq p(v)$ for $\forall v \in V$

1st useful application: Supporting hyperplane for a convex compact set $C \subset \mathbb{R}^n$.

2nd useful application: If A, B are convex, bounded, closed, with nonempty interior and $A \cap B = \emptyset$, then there exists a hyperplane $P_{u, \theta}$ such that:

$$\forall a \in A, \forall b \in B : \langle u, a \rangle - \theta > 0 > \langle u, b \rangle - \theta$$

8.2 Signed Measure

Definition 8.1 (Signed Measure): A signed measure on $(\Omega, \mathcal{F})$ is $\mu : \mathcal{F} \rightarrow [-\infty, \infty]$ for which:

1. $\mu(\emptyset) = 0$
2. μ takes at most one of: $-\infty, +\infty$
3. μ is σ -additive

Every signed measure $\mu = \mu^+ - \mu^-$ (Hahn–Jordan decomposition) and $\int f d\mu = \int f d\mu^+ - \int f d\mu^-$.

Theorem 8.2 (Riesz): If $L : C(I) \rightarrow \mathbb{R}$ is a bounded linear functional, then there exists a

finite signed measure μ on $(I, \mathcal{B}(I))$ such that:

$$L(f) = \int f d\mu \quad \text{for every } f \in C(I)$$

8.3 Cybenko's Theorem

Definition 8.2 (Sigmoid): We say that $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a **sigmoid**, if:

$$\lim_{x \rightarrow +\infty} \sigma(x) = 1, \quad \lim_{x \rightarrow -\infty} \sigma(x) = 0$$

Examples: $\text{sgn}(x)$, $\sigma(x) = \frac{1}{1+e^{-x}}$, $\text{ReLU}(x) - \text{ReLU}(x - 1)$.

Definition 8.3 (Discriminator): We say that $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a **discriminator**, if every finite signed measure μ for which:

$$\int \sigma(\langle x, y \rangle + \theta) d\mu = 0 \quad \forall y \in \mathbb{R}^d, \theta \in \mathbb{R}$$

satisfies $\int f d\mu = 0$ for every $f \in C(I)$.

$$G_\sigma = \left\{ f(x) = \sum_{i=1}^n d_i \sigma(\langle x, y_i \rangle + \theta_i) \mid y_i \in \mathbb{R}^d, \theta_i, d_i \in \mathbb{R}, n \in \mathbb{N} \right\}$$

Theorem 8.3 (Cybenko): Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous sigmoid. Then $\overline{G_\sigma} = C(I)$, i.e. for $\forall f \in C(I)$ and $\varepsilon > 0$ there exists $g \in G_\sigma$:

$$\max_{x \in I} |f(x) - g(x)| < \varepsilon$$

(G_σ is everywhere dense in $C(I)$).

Remark 8.4: Practical significance: Cybenko's theorem shows that a *single-layer* neural network with a continuous sigmoid is a *universal approximator* — it can approximate any continuous function on a compact set to arbitrary precision. It does **not** guarantee, however, how many neurons are needed (they may be exponentially many) and does **not** give a constructive algorithm for finding the weights. **Inaccuracy:** In the notes $\text{ReLU}(x) - \text{ReLU}(x - 1)$ is given as an example of a sigmoid — it is *not* a continuous sigmoid by definition (it does not tend to 1), but is a “bump” function; Cybenko's theorem requires a continuous sigmoid, but more general results (Hornik 1991) also cover ReLU.

See [Cyb89]: Theorem 2, pp. 308–310; K. Hornik, *Neural Networks* 4(2), 1991 (generalization to arbitrary continuous functions); [AB99]: ch. 6–7.

Proposition 8.5: If σ is a discriminator, then $\overline{G_\sigma} = C(I)$.

Proof. If $G_\sigma \neq C(I)$ (closed linear subspace), then for $f \in C(I) \setminus G_\sigma$, $d = \inf \|f - g\|_\infty > 0$. Define $L : G_\sigma \oplus \langle f \rangle \rightarrow \mathbb{R}$ by $L(g + df) = d$. By Hahn–Banach it extends to $L' : C(I) \rightarrow \mathbb{R}$. By Riesz’s theorem there exists a measure μ : $L'(h) = \int h d\mu$. For $h = \sigma(\langle \cdot, y \rangle + \theta) \in G_\sigma$: $\int h d\mu = 0$ by the discriminator property. But $\int f d\mu = d > 0$ — a contradiction. $\square$

Proposition 8.6: If $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous sigmoid, then σ is a continuous discriminator.

Proof. Let μ be a finite signed measure on $(I, \mathcal{B}(I))$ such that

$$\int_I \sigma(\langle x, y \rangle + \theta) d\mu(x) = 0 \quad \text{for every } y \in \mathbb{R}^d, \theta \in \mathbb{R}.$$

We need to show that $\int_I f d\mu = 0$ for every $f \in C(I)$.

Step 1: The measure of half-spaces and hyperplanes is zero.

For fixed $y \in \mathbb{R}^d$ and $\theta \in \mathbb{R}$ consider the sequence of continuous functions:

$$\lambda_n(x) = \sigma(n(\langle y, x \rangle + \theta)), \quad n \in \mathbb{N}.$$

Since σ is a sigmoid ($\lim_{t \rightarrow +\infty} \sigma(t) = 1$, $\lim_{t \rightarrow -\infty} \sigma(t) = 0$), we have pointwise:

$$\lim_{n \rightarrow \infty} \lambda_n(x) = \begin{cases} 1, & \langle y, x \rangle + \theta > 0, \\ \sigma(0), & \langle y, x \rangle + \theta = 0, \\ 0, & \langle y, x \rangle + \theta < 0. \end{cases}$$

The functions λ_n are bounded ($|\lambda_n| \leq \max_{t \in \mathbb{R}} |\sigma(t)|$ on the compact I), and by assumption $\int_I \lambda_n d\mu = 0$ (substituting $y \mapsto ny$, $\theta \mapsto n\theta$). By Lebesgue’s dominated convergence theorem:

$$\lim_{n \rightarrow \infty} \int_I \lambda_n d\mu = \int_I \lim_{n \rightarrow \infty} \lambda_n d\mu = \mu(H^+(y, \theta)) + \sigma(0) \mu(\Pi(y, \theta)) = 0,$$

where $H^+(y, \theta) = \{x \in I \mid \langle y, x \rangle + \theta > 0\}$ and $\Pi(y, \theta) = \{x \in I \mid \langle y, x \rangle + \theta = 0\}$.

Substituting $y \mapsto -y$, $\theta \mapsto -\theta$ (i.e. considering $H^-(y, \theta)$):

$$\mu(H^-(y, \theta)) + \sigma(0) \mu(\Pi(y, \theta)) = 0.$$

Since $H^+(y, \theta)$ and $H^-(y, \theta)$ are disjoint and $\mu(H^+) + \mu(\Pi) + \mu(H^-) = \mu(I)$ is finite, subtracting gives $\mu(H^+(y, \theta)) = \mu(H^-(y, \theta))$, and adding gives $\mu(\Pi(y, \theta)) = 0$. Therefore $\mu(H^+(y, \theta)) = 0$ and $\mu(H^-(y, \theta)) = 0$ for every y, θ .

Step 2: The measure of Borel strips is zero.

For $a < b$ the strip $B(y, a, b) = \{x \in I \mid a \leq \langle y, x \rangle \leq b\}$ can be represented as:

$$B(y, a, b) = I \setminus (H^+(y, -a) \cup H^-(y, -b)),$$

whence $\mu(B(y, a, b)) = \mu(I) - \mu(H^+) - \mu(H^-) - (\text{correction for hyperplanes}) = 0$.

More directly: the function $\mathbb{1}_{[a,b]}(\langle y, \cdot \rangle)$ is measurable and bounded. We approximate it uniformly by continuous functions of the form $\sigma(n(\langle y, x \rangle - a)) - \sigma(n(\langle y, x \rangle - b))$, for each of which the integral against μ is zero (by Step 1). Then $\int_I \mathbb{1}_{[a,b]}(\langle y, x \rangle) d\mu(x) = 0$.

Step 3: The measure annihilates trigonometric polynomials.

For $m \in \mathbb{Z}^d$ consider $\cos(2\pi\langle m, x \rangle)$. The function $t \mapsto \cos(2\pi t)$ is a limit (uniform) of linear combinations of interval indicators. Applying Step 2 and linearity:

$$\int_I \cos(2\pi\langle m, x \rangle) d\mu(x) = 0 \quad \text{and} \quad \int_I \sin(2\pi\langle m, x \rangle) d\mu(x) = 0.$$

Therefore $\int_I e^{2\pi i\langle m, x \rangle} d\mu(x) = 0$ for every $m \in \mathbb{Z}^d$.

Step 4: $\mu = 0$ on $C(I)$.

The trigonometric polynomials $\{\sum_{\text{finite}} c_m e^{2\pi i\langle m, x \rangle}\}$ are dense in $C(I)$ by the Stone–Weierstrass theorem. For arbitrary $f \in C(I)$ and $\varepsilon > 0$ there exists a trigonometric polynomial p with $\|f - p\|_\infty < \varepsilon$. Then:

$$\left| \int_I f d\mu \right| \leq \left| \int_I (f - p) d\mu \right| + \underbrace{\left| \int_I p d\mu \right|}_{=0} \leq \varepsilon |\mu|(I).$$

Since $\varepsilon > 0$ is arbitrary and $|\mu|(I) < \infty$, we get $\int_I f d\mu = 0$. □

9 Fourier Series

De Moivre's formula: $\cos \alpha + i \sin \alpha = e^{i\alpha}$.

$$\int_0^1 e^{2\pi i k \alpha} \cdot \overline{e^{2\pi i m \alpha}} d\alpha = \int_0^1 e^{2\pi i (k-m)\alpha} d\alpha = \begin{cases} 1, & k = m \\ 0, & k \neq m \end{cases}$$

$\Rightarrow \{e^{2\pi i k x}\}_{k \in \mathbb{Z}}$ form an orthonormal basis.

For $f : [0, 1] \rightarrow \mathbb{R}$ continuous/integrable:

$$c_f(m) = \int_0^1 f(x) e^{-2\pi i m x} dx = \langle f, e^{2\pi i m \cdot} \rangle$$

In $L^2([0, 1])$: $\int f^2(x) dx \geq \sum_{k \in \mathbb{Z}} |c_f(k)|^2$ (Parseval's inequality).

Corollary 9.1: If f is continuous: $\lim_{k \rightarrow \pm\infty} |c_f(k)| = 0$.

Theorem 9.2 (Parseval's Equality): For $f \in L^2([0, 1])$:

$$\int_0^1 f^2(x) dx = \sum_{k \in \mathbb{Z}} |c_k(f)|^2$$

Proposition 9.3: Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous and differentiable at $x_0 \in (0, 1)$. Then:

$$f(x_0) = \lim_{n \rightarrow \infty} \sum_{k=-n}^n c_k(f) e^{2\pi i k x_0}$$

Proof. **Step 1: Expressing the partial sum via the Dirichlet kernel.**

By definition of the Fourier coefficients:

$$\begin{aligned} S_n(f, x_0) &:= \sum_{k=-n}^n c_k(f) e^{2\pi i k x_0} = \sum_{k=-n}^n \left(\int_0^1 f(x) e^{-2\pi i k x} dx \right) e^{2\pi i k x_0} \\ &= \sum_{k=-n}^n \int_0^1 f(x) e^{-2\pi i k x} e^{2\pi i k x_0} dx \\ &= \int_0^1 f(x) \sum_{k=-n}^n e^{2\pi i k (x_0 - x)} dx. \end{aligned}$$

We compute the inner sum via a geometric series with $u = e^{2\pi i (x_0 - x)}$:

$$D_n(t) := \sum_{k=-n}^n e^{2\pi i k t} = e^{-2\pi i n t} \cdot \frac{e^{2\pi i (2n+1)t} - 1}{e^{2\pi i t} - 1} = \frac{e^{2\pi i (n+\frac{1}{2})t} - e^{-2\pi i (n+\frac{1}{2})t}}{e^{i\pi t} - e^{-i\pi t}} = \frac{\sin \pi (2n+1)t}{\sin \pi t},$$

called the **Dirichlet kernel**. Therefore:

$$S_n(f, x_0) = \int_0^1 f(x) D_n(x_0 - x) dx = \int_0^1 f(x) \frac{\sin \pi (2n+1)(x - x_0)}{\sin \pi (x - x_0)} dx.$$

Step 2: Normalization — $\int_0^1 D_n(t) dt = 1$.

We apply the formula for $f \equiv 1$: $c_k(1) = \int_0^1 e^{-2\pi i k x} dx = \mathbb{1}_{k=0}$, whence $S_n(1, x_0) = 1$ for every n and x_0 . Therefore:

$$\int_0^1 D_n(t) dt = 1.$$

Step 3: Reduction to a continuous auxiliary function.

We write the difference:

$$S_n(f, x_0) - f(x_0) = \int_0^1 (f(x) - f(x_0)) D_n(x_0 - x) dx = \int_0^1 \underbrace{\frac{f(x) - f(x_0)}{\sin \pi(x - x_0)}}_{=: g(x)} \cdot \sin \pi(2n+1)(x - x_0) dx.$$

Since f is differentiable at x_0 , the limit at x_0 is:

$$\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \frac{x - x_0}{\sin \pi(x - x_0)} = f'(x_0) \cdot \frac{1}{\pi}.$$

We set $g(x_0) = f'(x_0)/\pi$. Then $g : [0, 1] \rightarrow \mathbb{R}$ is continuous: the denominator $\sin \pi(x - x_0)$ vanishes only at x_0 (in $[0, 1]$), and there the limit from above exists and is finite.

Step 4: The Riemann–Lebesgue Lemma.

Lemma: For every $h \in L^1([0, 1])$:

$$\int_0^1 h(x) \sin(\omega x) dx \xrightarrow{\omega \rightarrow +\infty} 0.$$

Proof: For $h = \mathbb{1}_{[a,b]}$: $\int_a^b \sin(\omega x) dx = \frac{\cos(\omega a) - \cos(\omega b)}{\omega} \rightarrow 0$. By linearity it holds for step functions, and they are dense in L^1 : for $\varepsilon > 0$ there exists a step function φ with $\|h - \varphi\|_1 < \varepsilon$, whence:

$$\left| \int_0^1 h \sin \omega \right| \leq \|h - \varphi\|_1 + \left| \int_0^1 \varphi \sin \omega \right| \leq \varepsilon + o(1) \rightarrow \varepsilon.$$

Conclusion.

We apply the lemma with $h(x) = g(x) \cos(2n+1)\pi x_0 - g(x_0 - x + x_0)$ and $\omega = \pi(2n+1)$. More directly: $g \in C([0, 1]) \subset L^1([0, 1])$, whence:

$$S_n(f, x_0) - f(x_0) = \int_0^1 g(x) \sin(\pi(2n+1)(x - x_0)) dx \xrightarrow{n \rightarrow \infty} 0.$$

Therefore:

$$f(x_0) = \lim_{n \rightarrow \infty} \sum_{k=-n}^n c_k(f) e^{2\pi i k x_0}.$$

Corollary 9.4 (Absolute and Uniform Convergence for Smooth Functions): If $f : [0, 1] \rightarrow \mathbb{R}$ is of class C^2 , then the Fourier series converges absolutely and uniformly to f .

Proof. Integration by parts twice: $c_k(f'') = (2\pi i k)^2 c_k(f)$, whence $|c_k(f)| = |c_k(f'')|/(4\pi^2 k^2) \leq \|f''\|_\infty/(4\pi^2 k^2)$. Then $\sum_{k \in \mathbb{Z}} |c_k(f)| < \infty$ and by Weierstrass's M-test the series converges uniformly. $\square$

Corollary 9.5 (Riemann–Lebesgue Lemma): For every $f \in L^1([0, 1])$:

$$c_k(f) = \int_0^1 f(x) e^{-2\pi i k x} dx \xrightarrow{|k| \rightarrow \infty} 0.$$

Proof. From Step 4 (with $h = f \cdot \cos(2\pi k x)$ and $h = f \cdot \sin(2\pi k x)$): $\operatorname{Re}(c_k(f)) \rightarrow 0$ and $\operatorname{Im}(c_k(f)) \rightarrow 0$. □

□

Generalization in $\mathbb{R}^d$: For $I = [0, 1]^d$, $\vec{k} \in \mathbb{Z}^d$:

$$c_{\vec{k}}(f) = \int_I f(\vec{x}) e^{-2\pi i \langle \vec{k}, \vec{x} \rangle} d\vec{x}$$

$$f(\vec{x}) = \lim_{n \rightarrow \infty} \sum_{\vec{k} \in [-n, n]^d} c_{\vec{k}}(f) e^{2\pi i \langle \vec{k}, \vec{x} \rangle}$$

10 Kolmogorov–Arnold Theorem via ReLU Architecture

$f : I \rightarrow \mathbb{R}$, $I = [0, 1]^d$, f continuous, $(x_1, \dots, x_d)$.

ReLU:

$$\text{ReLU}(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Idea of “bit interleaving”:

We encode $x \in [0, 1]$ by binary expansion $x = 0.a_1^1 a_1^2 a_2^1 a_2^2 \dots$

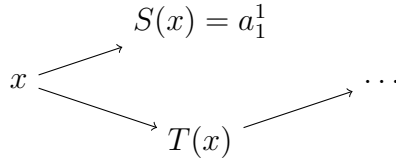
We define:

$$S(x) = 2^{-1} \text{ReLU}(x + 2^{-1}) - 2^{-1} \text{ReLU}(x - 2^{-1})$$

$$S(x) = \begin{cases} 1, & x > \frac{1}{2} + 2^{-r} \\ 0, & x < \frac{1}{2} - 2^{-r} \\ 2^{r-1}(x - \frac{1}{2}) + \frac{1}{2}, & x \in (\frac{1}{2} - 2^{-r}, \frac{1}{2} + 2^{-r}) \end{cases}$$

$a_1 = S(x)$; $T(x) = 2x - S(x)$ is $0.a_2^1 a_3^1 \dots$

Diagram (ReLU network for bit extraction):



Reduction:

$$g(\psi(x_1, \dots, x_d)) = f(x_1^{\leq r}, x_2^{\leq r}, \dots, x_d^{\leq r})$$

Since we are only interested in the first r bits of $x_1, \dots, x_d$, there are 2^{rd} distinct values.

g takes values at 2^{rd} points $0 = s_0 < s_1 < \dots \leq 1$. Linear interpolation (piecewise-linear approximation):

$$g(s) = g(s_0) + \sum_{i=0}^{2^{rd}-1} \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i} \text{ReLU}(s - s_i) - \sum_{i=0}^{2^{rd}-1} \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i} \text{ReLU}(s - s_{i+1})$$

Explanation (by analogy with $S(x)$):

Each linear segment $[s_i, s_{i+1}]$ with slope $m_i = \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i}$ is realized via:

$$m_i \cdot \text{ReLU}(s - s_i) - m_i \cdot \text{ReLU}(s - s_{i+1}) = \begin{cases} 0, & s < s_i \\ m_i(s - s_i), & s_i \leq s \leq s_{i+1} \\ m_i(s_{i+1} - s_i) = g(s_{i+1}) - g(s_i), & s > s_{i+1} \end{cases}$$

Summing over all segments and adding $g(s_0) = g(0)$, we obtain $g(s)$ for every $s \in [0, 1]$. Hence g is expressible by a finite number ($O(2^{rd})$) of ReLU functions — expressible via a ReLU network.

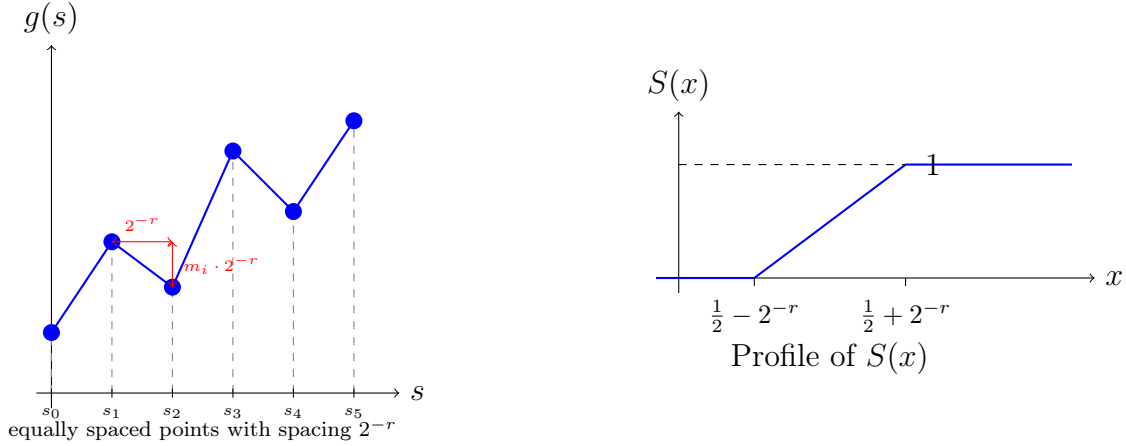
Remark (approximation with error 2^{-r}):

The real $x \in [0, 1]$ has an infinite binary representation; we truncate it to r bits, obtaining $x^{\leq r}$ with error $|x - x^{\leq r}| \leq 2^{-r}$. Since f is uniformly continuous on the compact $[0, 1]^d$, for every $\varepsilon > 0$ there exists r large enough such that:

$$\sup_{s \in [0, 1]} |g(s) - f(x_1, \dots, x_d)| < \varepsilon.$$

The realization is continuous by construction (linear combination of ReLU functions).

Visual representation:



See [AB99]: ch. 14 (approximation bounds for neural networks); Barron 1993 (approximation by sigmoids); Cybenko [Cyb89]; for risk optimization — [Vap99]: ch. 4–5.

10.1 Encoding $[0, 1]^d$ via Ternary Representation

Setup. We have $f : [0, 1]^d \rightarrow \mathbb{R}$ a continuous function and wish to approximate it with a ReLU neural network. The key idea is to encode the input $\bar{x} \in [0, 1]^d$ via the ternary (base 3) number system.

For $x \in [0, 1]$ we write the ternary expansion:

$$x = \sum_{k=1}^{\infty} a_k^x \cdot 3^{-k}, \quad a_k^x \in \{0, 1, 2\}, \quad \text{i.e. } x = [0. a_1^x a_2^x \dots]_{(3)}.$$

The function $\mathcal{X}_d$. For $x \in [0, 1]$ and $d \in \mathbb{N}$ we define:

$$\mathcal{X}_d(x) = \sum_{k=1}^{\infty} 2a_k^x \cdot 3^{-(k-1)d-1},$$

and for $K \in \mathbb{N}$ — its K -th partial approximation:

$$\mathcal{X}_d^{(K)}(x) = \sum_{k=1}^K 2a_k^x \cdot 3^{-(k-1)d-1}.$$

Remark 10.1: Intuition: $\mathcal{X}_d(x)$ “weights” the ternary digits of x with weights $3^{-(k-1)d-1}$. The factor d in the exponent ensures that when encoding d coordinates together the intervals do not overlap — each coordinate occupies its own “frequency band” in the ternary expansion.

Definition 10.1 (The function $\bar{e}$): We define $\bar{e} : [0, 1]^d \rightarrow \mathcal{Y}$, where:

$$\bar{e}(x_1, \dots, x_d) = \sum_{j=1}^d \mathcal{X}_d(x_j) \cdot 3^{-(j-1)},$$

and its K -th approximation:

$$\bar{e}^{(K)}(x_1, \dots, x_d) = \sum_{j=1}^d \mathcal{X}_d^{(K)}(x_j) \cdot 3^{-(j-1)}.$$

Remark 10.2: $\bar{e}$ gives a **bijection** from $[0, 1]^d$ to $\mathcal{Y}$, and $\bar{e}^{(K)}$ gives an approximation to $\bar{e}$. Visually: $\bar{e}$ “interleaves” the ternary expansions of d coordinates into a single real number in such a way that distinct points are sent to distinct images.

10.2 Separation Property of $\bar{e}$

Proposition 10.3 (Lower Bound for $|\bar{e}(\bar{x}) - \bar{e}(\bar{y})|$): If $\bar{x}, \bar{y} \in [0, 1]^d$ and $|x_i - y_i| > 2^{-j}$ for some i , then:

$$|\bar{e}(\bar{x}) - \bar{e}(\bar{y})| \geq 3^{-d(j-1)-i} \geq 3^{-dj}.$$

Proof. Idea: If the points are far apart in $[0, 1]^d$, then their images are far apart in $\mathcal{Y}$.

Step 1: Reduction to ternary digits. If $|x_i - y_i| > 2^{-j}$, then $(a_1^{x_i}, \dots, a_j^{x_i}) \neq (a_1^{y_i}, \dots, a_j^{y_i})$, because otherwise:

$$|x_i - y_i| \leq \sum_{s=j+1}^{\infty} 2 \cdot 3^{-s} = 2 \cdot 3^{-(j+1)} \cdot \frac{1}{1 - 3^{-1}} = 2^{-j} \cdot \frac{1}{2}.$$

More precisely: let $a_t^{x_i} \neq a_t^{y_i}$ and t be the minimal such index. Then:

$$\mathcal{X}_d(x_i) - \mathcal{X}_d(y_i) \geq 2 \cdot 3^{-(t-1)d-1} - \sum_{s=t+1}^{\infty} 2 \cdot 3^{-(s-1)d-1} = 2 \cdot 3^{-(t-1)d-1} - 2 \cdot 3^{-(t-1)d-1} \cdot \frac{3^{-d}}{1 - 3^{-d}}.$$

Since $\sum_{s=t}^{\infty} 3^{-st-1} = 3^{-dt-1} \cdot \frac{1}{1 - 3^{-d}}$, we obtain:

$$\mathcal{X}_d(x_i) - \mathcal{X}_d(y_i) \geq 2 \cdot 3^{-(t-1)d-1} \left[1 - \frac{3^{-d}}{1 - 3^{-d}} \right] = 2 \cdot 3^{-(t-1)d-1} \cdot \frac{1 - 2 \cdot 3^{-d}}{1 - 3^{-d}}.$$

Step 2: Difference in $\bar{e}$. The ternary representations of $\bar{e}(\bar{x})$ and $\bar{e}(\bar{y})$ first differ at position $\leq (t-1)d + i$. Therefore:

$$|\bar{e}(\bar{x}) - \bar{e}(\bar{y})| \geq 2 \cdot 3^t - 2 \sum_{s=t+1}^{\infty} 3^{-s} = 2 \cdot 3^t - 2 \cdot 3^t \cdot \frac{1/3}{1 - 1/3} = 2 \cdot 3^t \left(1 - \frac{1/3}{1 - 1/3} \right) = 2 \cdot 3^t \cdot \frac{1}{2} = 3^t \geq 3^{d(j-1)-i}.$$

Since $t \leq (t-1)d + i \leq (j-1)d + i$, we obtain:

$$|\bar{e}(\bar{x}) - \bar{e}(\bar{y})| \geq 3^{-d(j-1)-i} \geq 3^{-dj}. \quad \square$$

Corollary: If the points are far apart in $[0, 1]^d$, then their images are far apart. That is, Γ_x and Γ_y first differ at position $\leq (t-1)d + i$. $\square$

10.3 The Function g and Its Approximations g^k

Definition 10.2 (The Function g and Approximations g^k): For $f : [0, 1]^d \rightarrow \mathbb{R}$ we define $g : \mathcal{Y} \rightarrow \mathbb{R}$ by:

$$g(x) = f(\bar{e}^{-1}(x)).$$

For $K \in \mathbb{N}$ we define $g^k : \mathcal{Y} \rightarrow \mathbb{R}$ and $g_k : \mathcal{Y} \rightarrow \mathbb{R}$:

$$g^k(x) = g(\bar{e}^{(k)}(\bar{e}^{-1}(x))), \quad g^k(x) = f(\bar{e}^{-1}(x)).$$

Remark 10.4: Intuition: g “transfers” the function f from $[0, 1]^d$ to the space $\mathcal{Y}$ via the encoding $\bar{e}$. The function g^k uses only k ternary digits for approximation — it is a step function with accuracy $\sim 3^{-kd}$.

10.4 Theorem on Lipschitz Approximability via g

Theorem 10.5 (Modulus of Continuity of g): If $f : [0, 1]^d \rightarrow \mathbb{R}$ is L -Lipschitz in the sense that $\forall \bar{x}, \bar{y} \in [0, 1]^d$:

$$|f(\bar{x}) - f(\bar{y})| \leq L \|\bar{x} - \bar{y}\|_\infty,$$

then for $x, y \in \mathcal{Y}$:

$$|g(x) - g(y)| \leq 2L |x - y|^{1/d \log_2 3}.$$

If f is continuous (but not necessarily Lipschitz), then g is continuous with modulus of continuity $1/(d \log_2 3)$, i.e. Hölder-continuous with exponent $1/d \log_2 3$.

Proof. Idea: If x, y are close in $\mathcal{Y}$, then $\bar{e}^{-1}(x)$ and $\bar{e}^{-1}(y)$ are close in $[0, 1]^d$, whence $g(x) = f(\bar{e}^{-1}(x))$ and $g(y)$ are close by the Lipschitz property of f .

Step 1. Let $x, y \in [0, 1]$:

$$|g(x) - g(y)| = |f(\bar{e}^{-1}(x)) - f(\bar{e}^{-1}(y))| \stackrel{L\text{-Lipsch.}}{\leq} L \|\bar{e}^{-1}(x) - \bar{e}^{-1}(y)\|_\infty.$$

Step 2. Let $j \in \mathbb{N}$ be such that $\|\bar{e}^{-1}(x) - \bar{e}^{-1}(y)\|_\infty \in [2^{-j}, 2^{-(j-1)})$. From Proposition 10.3:

$$\|\bar{e}^{-1}(x) - \bar{e}^{-1}(y)\|_\infty \geq 3^{-jd} = (2^{-j})^{\log_2 3 \cdot d}.$$

Therefore $2^{-j} \leq \|\dots\|_\infty^{1/(\log_2 3 \cdot d)}$.

Step 3. Now:

$$\|\bar{e}^{-1}(x) - \bar{e}^{-1}(y)\|_\infty \leq 2^{-(j-1)} = 2 \cdot 2^{-j} = 2 \cdot (3^{-jd})^{1/(\log_2 3 \cdot d)} \leq 2 |x - y|^{1/(d \log_2 3)},$$

whence $|g(x) - g(y)| \leq 2L |x - y|^{1/(d \log_2 3)}$. $\square$

10.5 Approximation of f via g^k

Theorem 10.6 (Approximation by g^k): If $f : [0, 1]^d \rightarrow \mathbb{R}$ is L -Lipschitz, then:

$$|f(\bar{x}) - g_k(\bar{e}^{(k)}(\bar{x}))| \leq 2L \cdot 2^{-k}.$$

Proof. Since $f(\bar{x}) = g(\bar{e}(\bar{x}))$:

$$|f(\bar{x}) - g(\bar{e}^{(k)}(\bar{x}))| = |g(\bar{e}(\bar{x})) - g(\bar{e}^{(k)}(\bar{x}))| \leq 2L |\bar{e}(\bar{x}) - \bar{e}^{(k)}(\bar{x})|^{1/(d \log_2 3)}.$$

The functions $\bar{e}(\bar{x})$ and $\bar{e}^{(k)}(\bar{x})$ agree in their first Kd digits after the decimal point in the ternary number system (only from position Kd onwards). Therefore:

$$|\bar{e}(\bar{x}) - \bar{e}^{(k)}(\bar{x})| \leq \sum_{j=Kd+1}^{\infty} 2 \cdot 3^{-j} = 3^{-Kd} \cdot \frac{2}{1 - 1/3} = 3^{-Kd},$$

whence:

$$|f(\bar{x}) - g(\bar{e}^{(k)}(\bar{x}))| \leq 2L (3^{-Kd})^{1/(d \log_2 3)} = 2L \cdot 2^{-K}.$$

It remains to note that $\bar{e}^{(k)}(\bar{x}) = \bar{e}^{(k)} \bar{e}^{-1} \bar{e}^{(k)}(\bar{x})$, i.e. $g(\bar{e}^{(k)}(\bar{x})) = g_k(\bar{e}^{(k)}(\bar{x}))$. □

10.6 ReLU Network for Lipschitz Functions

Definition 10.3 (ReLU Network of Type $(l, d_1, d_2, \dots, d_l)$): An acyclic neural network is a **ReLU network of type** $(l; d_1, d_2, \dots, d_l)$ if it has l hidden layers with $d_1, \dots, d_l$ neurons respectively and the activations are of ReLU type.

Theorem 10.7 (ReLU Approximation of Lipschitz Functions): Let $p \in [1, \infty)$ and $f : [0, 1]^d \rightarrow \mathbb{R}$ be L -Lipschitz: $|f(\bar{x}) - f(\bar{y})| \leq L \|\bar{x} - \bar{y}\|_{\infty}$.

Then for every $k \in \mathbb{N}$ there exists a ReLU network with $2k + 3$ hidden layers of type:

$$(2k + 3; \underbrace{4d, 4d, \dots, 4d}_{2k}, d, 1, 2^{kd} + 1)$$

with function $\tilde{f} : [0, 1]^d \rightarrow \mathbb{R}$, such that:

$$\|f - \tilde{f}\|_{p, \lambda([0, 1]^d)} \leq 2(L + \|f\|_{\infty}) \cdot 2^{-k}.$$

The parameters of the network are $\leq 2(Kd \|f\|_{\infty}) \cdot 2^{k(d \vee p)}$.

Remark 10.8: Intuition and significance:

- The theorem says that every Lipschitz function of d variables can be approximated to arbitrary precision 2^{-k} by a ReLU network with $O(k)$ layers.
- The depth $2k + 3$ grows *linearly* with the desired precision k .
- The width $4d$ of the hidden layers depends only on the input dimension d — not on k !

- The last wide layer $2^{kd} + 1$ encodes all 3^{kd} cells of the ternary partition, hence grows with k . This is precisely the main cost of the approximation.
- **Application in ML:** gives a constructive upper bound on the number of parameters needed for good generalization.

See [AB99]: ch. 14; Barron (1993) for the related result for sigmoids.

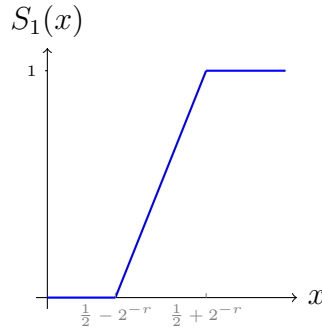
Proof. **Construction of the network.**

Step 1: The functions S_r and T_r .

For $x \in [0, 1]$ and parameter $r \in \mathbb{N}$ we define:

$$S_1(x) = 2^r \left(x - \frac{1}{2} + 2^{-r}\right)_+ - 2^r \left(x - \frac{1}{2} - 2^{-r}\right)_+, \quad T_1(x) = 2x,$$

$$S_{j+1}(x) = S_1(T_j(x) - S_j(x)), \quad T_{j+1}(x) = T_1(T_j(x) - S_j(x)).$$



Properties of S_r : For $x \notin [\frac{1}{2} - 2^{-r}, \frac{1}{2} + 2^{-r}]$: $S_1(x) = a_1^x$, $T_1(x) = a_1^x \cdot a_2^x \dots$ (extracts the ternary digits). $T_1(x) - S_1(x)$ corresponds to a left shift by 1 position and discarding a_1 (analogous to a pop from an infinite queue).

The measure $\lambda\left(\left(\frac{1}{2} - 2^{-r}, \frac{1}{2} + 2^{-r}\right)\right) = 2^{-r}$.

Step 2: Encoding d coordinates.

We define:

$$\mathcal{X}_d^{(k)}(x) = \sum_{s=1}^k 2 \cdot 3^{-(s-1)d-1} \cdot a_s^x,$$

$$\Phi_j(x) = \sum_{s=1}^k 2 \cdot 3^{-(sd-1)-s} \cdot a_s^x \quad (\text{partial sum}),$$

$$S_S(x) = \Phi_{j-1}(x) + 2 \cdot 3^{d(j-1)-1} \cdot S_j(x),$$

$$\bar{e}^{(k)}(x_1, \dots, x_d) = \sum_{i=1}^d 3^{-(i-1)} \mathcal{X}_d^{(k)}(x_i), \quad \prod_{i=1}^d [e_i 2^k, (l_i + 1) 2^k], \quad 0 \leq l_i < 2^k.$$

Step 3: Defining $\tilde{f}_0$.

Let $(s_j)_{j=0}^{2^{kd}-1}$ be all points in $[0, 1]$ of the form $\sum_{j=1}^{kd} 2a_j 3^{-j}$, $a_j \in \{0, 1\}$, ordered: $s_0 = 0 < s_1 < \dots < s_{2^{kd}-1}$.

We define the linear interpolant:

$$\tilde{f}_0(x) = g(s_0) + \sum_{i=0}^{2^{kd}-2} \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i} \left[(x - s_i)_+ - (x - s_{i+1})_+ \right],$$

where $g = f \circ \bar{e}^{-1}$.

Then for $s_i \leq x \leq s_{i+1}$:

$$\begin{aligned} \tilde{f}_0(x) &= g(s_0) + \sum_{j=0}^{i-1} (g(s_{j+1}) - g(s_j)) + \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i} (x - s_i) \\ &= g(s_i) + \frac{g(s_{i+1}) - g(s_i)}{s_{i+1} - s_i} (x - s_i) \\ &= g(s_i) \cdot \frac{s_{i+1} - x}{s_{i+1} - s_i} + g(s_{i+1}) \cdot \frac{x - s_i}{s_{i+1} - s_i}. \end{aligned}$$

Interpolation error:

$$\begin{aligned} |\tilde{f}_0(x) - f(\bar{e}^{-1}(x))| &= |g(s_i) \frac{s_{i+1}-x}{s_{i+1}-s_i} + g(s_{i+1}) \frac{x-s_i}{s_{i+1}-s_i} - f(\bar{e}^{-1}(x))| \\ &\leq \frac{s_{i+1}-x}{s_{i+1}-s_i} |g(s_i) - f(\bar{e}^{-1}(x))| + \frac{x-s_i}{s_{i+1}-s_i} |g(s_{i+1}) - f(\bar{e}^{-1}(x))| \\ &\leq \frac{s_{i+1}-x}{s_{i+1}-s_i} \cdot 2L \cdot 2^{-k} + \frac{x-s_i}{s_{i+1}-s_i} \cdot 2L \cdot 2^{-k} = 2L \cdot 2^{-k}. \end{aligned}$$

(the last 2 layers of the network realize each **specific** code of $\tilde{f}_0$).

Step 4: Final network.

The function $\tilde{f}(x_1, \dots, x_d) = \tilde{f}_0\left(\sum_{j=1}^d 3^{-(j-1)} \Phi_k(x_j)\right)$ is the function realized by the ReLU network. For it:

1. With probability $\geq 1 - 2^{-pk}$: $\Phi_k(x_i) = \mathcal{X}_d^{(k)}(x_i)$ for every i . On this event:

$$|\tilde{f}(x_1, \dots, x_d) - f(x_1, \dots, x_d)| = |\tilde{f}_0(\bar{e}^{(k)}(x_1, \dots, x_d)) - f(x_1, \dots, x_d)| = |g_k(\bar{e}^{(k)}(x_1, \dots, x_d)) - f(\dots)| \leq 2^{-k}.$$

2. On the set where $\Phi_k(x_i) \neq \mathcal{X}_d^{(k)}(x_i)$ for some i , we obtain a convex combination of 2 values $g_k(\bar{e}^{(k)}(\bar{y})) = f(\bar{y})$ and $g_k(\bar{e}^{(k)}(\bar{z})) = f(\bar{z})$. Each of them in absolute value is $\leq \|f\|_\infty$, whence:

$$|\tilde{f}(x_1, \dots, x_d) - f(x_1, \dots, x_d)| \leq 2\|f\|_\infty.$$

3. Let $B = \{\bar{x} \in [0, 1]^d \mid \Phi_k(x_i) \neq \mathcal{X}_d^{(k)}(x_i) \text{ for some } i \leq d\}$:

- $\lambda(B) \leq 2^{-pk}$
- $\bar{x} \in B$: $|f(\bar{x}) - \tilde{f}(\bar{x})| \leq 2\|f\|_\infty$
- $\bar{x} \notin B$: $|f(\bar{x}) - \tilde{f}(\bar{x})| \leq 2L \cdot 2^{-k}$

Therefore:

$$\begin{aligned}
\|f - \tilde{f}\|_{p, \lambda([0,1]^d)}^p &= \int_{[0,1]^d} |f(\bar{x}) - \tilde{f}(\bar{x})|^p d\bar{x} \\
&= \int_B |f - \tilde{f}|^p d\bar{x} + \int_{B^c} |f - \tilde{f}|^p d\bar{x} \\
&\leq \lambda(B)(2\|f\|_\infty)^p + (2L \cdot 2^{-k})^p \\
&\leq 2^{-pk}(2\|f\|_\infty)^p + (2L)^p \cdot 2^{-pk} \\
&= 2^{-pk} ((2\|f\|_\infty)^p + (2L)^p) \\
&\leq 2^{-pk} \cdot (2\|f\|_\infty + 2L)^p.
\end{aligned}$$

Taking the p -th root: $\|f - \tilde{f}\|_p \leq 2^{-k}(2\|f\|_\infty + 2L) = 2(L + \|f\|_\infty) \cdot 2^{-k}$.

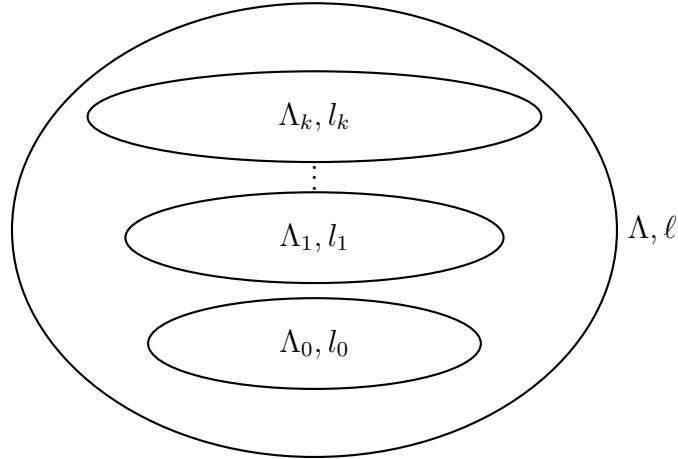
Remark 10.9: The set B (of appropriate measure) is not related to the distribution of the data. When we have coordinates $G(\frac{1}{2} - \frac{\varepsilon}{15})$ which are $\leq l < 2^r$. The bound via the continuity of f remains.

□

11 Risk Optimization: Hierarchy of Models

Setup. We have:

$$X = \mathbb{R}^{d_{in}}, \quad Y = \mathbb{R}^{d_{out}}, \quad \Lambda = \mathbb{R}^{d_\Lambda}, \quad h : \Lambda \times X \rightarrow Y, \quad \ell : Y \times Y \rightarrow [0, \infty), \quad (x, y) \sim P.$$



We structure (Λ, ℓ) in a hierarchy $\tilde{\Lambda}_k = (\Lambda_k, l \wedge B_k)$, such that:

1. $\Lambda_k \subseteq \Lambda_{k+1}$ (increasing sequence of models)
2. $B_k \nearrow \infty$ (the loss bound grows)
3. $\bigcup_{k=0}^\infty \Lambda_k$ is dense in Λ and $\ell(h(\lambda, x), y)$ is continuous in λ

Definition 11.1 (Level Selection Rule): A **rule** is a function $r : \mathbb{N} \rightarrow \mathbb{N}$, $m \mapsto r(m)$ — the level in the hierarchy.

Remark 11.1: Intuition: The goal is to optimize the risk $\rho(\lambda) = \mathbb{E} \ell(h(\lambda, x), y)$. Since Λ may be infinite (a neural network with millions of parameters), direct optimization is impossible. Instead:

- With little data (m small) — we use a simple model $\Lambda_{r(m)}$ with small VC-dimension $d_{r(m)}$.
- With much data — we increase expressibility by transitioning to Λ_k with larger k .
- The rule $r(m)$ balances the approximation error (how well Λ_k can approximate λ^*) and the generalization error (how well $\hat{\lambda}^{(k)}$ approximates $\lambda_{(k)}^*$ from data).

See [Vap99]: ch. 4–5 (structural risk minimization, SRM); [AB99]: ch. 11.

11.1 Convergence Theorem for Optimal Choice of $r(m)$

Theorem 11.2 (Risk Convergence under Rule $r(m)$): Let $r : \mathbb{N} \rightarrow \mathbb{N}$ be a rule for which:

- a) $r(m) \xrightarrow{m \rightarrow \infty} \infty$
- b) $\frac{B_{r(m)}^2 d_{r(m)} \log m}{m} \xrightarrow{m \rightarrow \infty} 0$ or $B_{r(m)} R_m(\Lambda_{r(m)}) \xrightarrow{m \rightarrow \infty} 0$

If $V(m) = \rho(\lambda^{r(m)}) - \rho(\lambda^*) + S(m)$, where:

$$S(m) = B_{r(m)} \sqrt{\frac{d_{r(m)} \ln(m+1)}{m}} \quad \text{or} \quad S(m) = B_{r(m)} R_m(\Lambda_{r(m)}),$$

then:

$$\mathbb{P} \left(\limsup_{m \rightarrow \infty} V^{-1}(m) |\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^*)| < \infty \right) = 1.$$

Remark 11.3: Interpretation: The theorem says that with probability 1 the generalization error $|\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^*)|$ grows *no faster* than $V(m)$.

$V(m) \rightarrow 0$ and therefore $V^{-1}(m) \rightarrow +\infty$.

With appropriate choice of $r(m)$: $\rho(\hat{\lambda}^{r(m)}) \rightarrow \rho(\lambda^*)$ almost surely.

That is, the algorithm is **asymptotically optimal**: with much data we find a model almost as good as the theoretically optimal λ^* .

Proof. Step 1: Bound on $\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^{r(m)})$.

We consider the case with the (B_k, d_k) -parametrization of Λ_k (i.e. l -oriented). From the density of $\bigcup \Lambda_k$ in Λ and the continuity of $\ell(h(\lambda, \cdot), \cdot)$ in λ , there exists $\lambda_k \in \Lambda_k$: $\lambda_k \rightarrow \lambda^*$ and $\rho(\lambda_k) \xrightarrow{k \rightarrow \infty} \rho(\lambda^*)$.

Therefore: $\rho(\lambda^{r(m)}) \xrightarrow{m \rightarrow \infty} \rho(\lambda^*)$.

Step 2: Probabilistic bound.

The goal is to control the probability that $\rho(\hat{\lambda}^{r(m)})$ is far from $\rho(\lambda^{r(m)})$.

Let $S(m) = \frac{1}{m\delta}$. Then we know that:

$$\mathbb{P}(A_m) < \delta(m) = \frac{1}{m^2},$$

$$\text{where } A_m = \left\{ \bar{z} \in Z^m \mid |\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^{r(m)})| > B_{r(m)} \cdot 32 \sqrt{\frac{d_{r(m)}(\ln \frac{m}{d_{r(m)}} + 1) + \ln \frac{2}{\delta(m)}}{m}} \right\}.$$

The choice $\delta(m) = \frac{1}{m^2}$ gives $\ln \frac{2}{\delta} = 2 \ln m + \ln 2 \sim 2 \ln m$.

Step 3: Almost sure convergence.

Let $\bar{z}^* = \{z_n\}_{n \in \mathbb{N}}$ be an infinite sequence with $z_n \in \mathcal{Z}$ and $\bar{z}(m) = (z_1, \dots, z_m) \in \mathcal{Z}^m$.

If there exists m_0 : for $m > m_0$, $\bar{z}(m) \notin A_m$, then:

$$\limsup_{m \rightarrow \infty} \frac{\rho(\hat{\lambda}^{r(m)}) - \rho^{r(m)}}{V(m)} \leq \limsup_{m \rightarrow \infty} \frac{32 B_{r(m)} \sqrt{\frac{d_{r(m)} \ln \frac{m}{d_{r(m)}} + 2 \ln 2m}{m}}}{V(m)} \leq 34.$$

It remains to show that with probability 1 there exists $m_0 \in \mathbb{N}$: for $m > m_0$, $\bar{z}(m) \notin A_m$. This is a property of $\bigcup_{m_0} \bigcap_{m > m_0} A_m^c$. It suffices to show:

$$\mathbb{P}\left(\bigcap_{m_0} \bigcup_{m > m_0} A_m\right) = 0.$$

Let $A'_m = \bigcup_{n \geq m} A_n$. Clearly $A'_{m+1} \subseteq A'_m$ and $\mathbb{P}(\bigcap A'_m) = \lim_{m \rightarrow \infty} \mathbb{P}(A'_m)$.

But:

$$\mathbb{P}(A'_m) \leq \sum_{n \geq m} \mathbb{P}(A_n) \leq \sum_{n \geq m} \delta(n) = \sum_{n \geq m} \frac{1}{n^2} \xrightarrow{m \rightarrow \infty} 0,$$

whence $\lim_{m \rightarrow \infty} \mathbb{P}(A'_m) = 0$, i.e. the event $\bigcup_{m_0} \bigcap_{m > m_0} A_m^c$ occurs almost surely.

Step 4: Final bound.

On $\bigcup_{m_0} \bigcap_{m > m_0} A_m^c$ we have:

$$\begin{aligned} \limsup_{m \rightarrow \infty} \frac{\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^*)}{V(m)} &= \limsup_{m \rightarrow \infty} \left[\frac{\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^{r(m)})}{V(m)} + \frac{\rho(\lambda^{r(m)}) - \rho(\lambda^*)}{V(m)} \right] \\ &\leq \limsup_{m \rightarrow \infty} \frac{\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^{r(m)})}{V(m)} + 1 \leq 34 + 1 = 35. \end{aligned}$$

Therefore $\mathbb{P}\left(\limsup_{m \rightarrow \infty} V^{-1}(m) |\rho(\hat{\lambda}^{r(m)}) - \rho(\lambda^*)| < \infty\right) = 1$. □ □

Remark 11.4 (Bound on $R_m(\Lambda_k)$ via Lipschitz continuity): Under the Rademacher approach:

$$R_m(\Lambda_k) \leq \frac{L_p(\ell \wedge B_k)}{\sqrt{m}} \sup_{\lambda \in \Lambda_k} \|\lambda\|_2,$$

where L_p is the Lipschitz constant of ℓ . This allows explicit computation of the convergence rate for a given architecture. See [AB99]: ch. 28–29; [LO19] for adaptive step sizes in SGD optimization.

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